Project Title : 01

Android App Development using GenAI- Shishu-Sneh (Healthcare)

1. The Problem Statement:

New mothers in villages often stop "Exclusive Breastfeeding" early because of myths or return-to-work pressure. They don't have a way to track the baby's "Weight Gain" or "Vaccination Milestones" once they leave the hospital.

2. Detailed Description (The Vision)

Shishu-Sneh is a "Baby’s First Year Guide." It acts as a digital elder. It provides weekly "Developmental Milestones" (e.g., "Baby should smile now"), a simple "Weight Tracker," and an "Immunization Calendar." It sends alerts for the "Polio Drops" and other vital shots.

3. App Usage & User Flow

- **Growth Chart:** Enter weight/height once a month -> See a simple trend line.
- **Vaccination Alert:** Upcoming shots with the date and "Disease it prevents."
- **Feeding Guide:** Daily tips on nutrition for the mother and the baby.
- **Milestone Checklist:** "Is the baby holding their head up?" (Yes/No).

4. Technical Implementation & Hints

- **Visualization:** Use MPAndroidChart for the growth trend line.
- **Alarms:** WorkManager for vaccination reminders.
- **Database:** Room DB to keep the baby's health record private.

5. Impact Goals

- **Child Survival:** Reducing infant mortality through timely vaccinations.
- **Stunting Prevention:** Ensuring parents understand proper nutrition and growth.
- **Health Equity:** Providing expert medical guidance in the local language.

6. Success Criteria for Students

- The "Growth Chart" must show a clear progress over 6 months.
- Reminders must be set automatically based on the baby's date of birth.
- The UI must be soft, clean, and mother-friendly.

Project Title : 02

Android App Development using GenAI- Shaale-Vikas (Education/Infrastructure)

1. The Problem Statement:

Rural schools have "Repair Needs" (e.g., leaking roof, broken desk). By the time the government fund arrives, the damage is worse. Alumni of the school (Old Students) want to help, but they don't know what the "Current Priority" of the school is.

2. Detailed Description (The Vision)

Shaale-Vikas is a "School-Alumni Bridge." It is a transparency tool for school development. The Headmaster lists "Micro-Needs" (e.g., "Need 5 sets of paints for the wall"). Alumni living in cities can see these needs and "Pledge" to fulfill them—either by donating the items or the funds. It turns the alumni network into a "Support System."

3. App Usage & User Flow

- **Needs Dashboard:** A list of items/repairs needed with a "Cost Estimate."
- **Progress Bar:** "60% of funds collected for the Toilet Repair."
- **Donor Hall of Fame:** A list of people who have helped.
- **Impact Photos:** "Before and After" photos of the completed work.

4. Technical Implementation & Hints

- **UI:** Use Progress Bars and CardViews.
- **Real-time:** Use Firebase to update the collection status instantly.
- **Media:** Use ImagePicker for the headmaster to upload "Need" photos.

5. Impact Goals

- **Community Ownership:** Empowering citizens to take care of their local schools.
- **Education Quality:** Ensuring infrastructure gaps don't affect learning.
- **Transparency:** Building trust through "Before/After" visual proof.

6. Success Criteria for Students

- The "Needs Dashboard" must be easily editable by the "Admin."
- The "Pledge" button must not involve actual money (Simulate the commitment).
- The UI must be professional and inspiring.

Project Title :03

Android App Development using GenAI-
Namma-Vastra (Self-Employment)

1. The Problem Statement:

Ilkal and Molakalmuru weavers create world-class sarees, but they only sell to "Wholesalers" who take a huge cut. The weaver doesn't know what "Modern Colors" (e.g., Pastels, Greys) are trending in the city, so they keep making the same traditional combinations that are slow to sell.

2. Detailed Description (The Vision)

Namma-Vastra is a "Weaver-to-Market Trend Guide." It provides weavers with "City Fashion Insights." The app shows "Trending Colors of the Month" and "New Pattern Ideas" curated for handloom. It also allows the weaver to post a photo of their "Loom-Fresh" saree to a community gallery for urban boutiques to see.

3. App Usage & User Flow

- **Trend Board:** Photos of modern sarees and color palettes.
- **Loom Gallery:** Upload a photo of a finished saree to a public catalog.
- **Material Costing:** Enter Silk/Cotton price -> App suggests a "Fair Retail Price."
- **Weaver's Story:** A section to document the history of the Ilkal/Molakalmuru craft.

4. Technical Implementation & Hints

- **UI:** Use Coil for high-quality image loading.
- **Database:** Firebase Firestore for the public catalog.
- **Logic:** A simple "Price Calculator" formula.

5. Impact Goals

- **Artisanal Income:** Increasing the profit margin for the actual creator.
- **Handloom Revival:** Keeping traditional weaving alive by making it "Fashionable."
- **Digital Market:** Moving the "Loom" directly into the "Global Marketplace.

6. Success Criteria for Students

- The "Loom Gallery" must allow users to "Inquire" via WhatsApp.
- The "Trend Board" must be updated with new images (simulated).
- The UI must be elegant and focus on the beauty of the fabric.

Project Title :04

Android App Development using GenAI- Grama-Yatri (Mobility)

1. The Problem Statement:

Wait times for the "Village Bus" are unpredictable. Passengers (mostly labor and students) often miss the bus by 2 minutes and have to wait for 3 hours for the next one. They need a "Community-Powered" timetable that is updated in real-time.

2. Detailed Description (The Vision)

Grama-Yatri is a "Shared Transit Log." It is a crowdsourced bus tracker. If a passenger sees the bus arrive at the first stop, they "Ping" the app. The app then calculates the "Expected Time" at all subsequent villages on that route. It's a "Live Timetable" created by the passengers themselves.

3. App Usage & User Flow

- **Route View:** A list of all villages on the route.
- **Bus Ping:** "I am on the bus" or "Bus just passed me."
- **Live ETA:** "Bus will reach your village in ~15 mins."
- **Alert: A notification:** "The morning bus is canceled today" (User-reported).

4. Technical Implementation & Hints

- **Real-time:** Use Firebase Realtime Database for instant pings.
- **Logic:** Use a List of stops with "Average Travel Time" between them.
- **UI:** Use a Vertical Stepper or Timeline View for the route.

5. Impact Goals

- **Efficient Mobility:** Reducing wasted man-hours at bus stops.
- **Economic Connectivity:** Helping rural workers reach city jobs on time.
- **Social Equity:** Making rural transport as predictable as city metros.

6. Success Criteria for Students

- The "Live ETA" must update for all users on the route instantly.
- The app must show the "Source" of the report (e.g., "Reported by Ravi").
- The UI must be extremely low-data and fast.

Project Title :05

Android App Development using GenAI- Namma-Kathey (National Pride)

1. The Problem Statement:

Typically every district has a "Local Legend" or "Historical Hero" (e.g., Sangolli Rayanna, Kittur Chennamma). Children know about global superheroes but don't know the "Bravery Stories" from their own district.

2. Detailed Description (The Vision)

Namma-Kathey is a "Regional Hero Storybook." It is an interactive history app for children. It tells stories of local freedom fighters, poets, and social reformers from every district of Karnataka using simple language and illustrations. It builds "Local Pride" and "Character" through storytelling.

3. App Usage & User Flow

- **District Map:** Tap a district to see its "Heroes."
- **Illustrated Story:** A swipe-based storybook in Kannada/English.
- **Hero Quiz:** Answer 3 questions to earn a "Heritage Badge."
- **Statue Finder:** Location of the nearest memorial/statue of that hero.

4. Technical Implementation & Hints

- **UI:** Use ViewPager2 for the swipeable storybook experience.
- **Media:** Use Text-To-Speech to read stories aloud.
- **Database:** Store stories and images in a local JSON file.

5. Impact Goals

- **Patriotism:** Building a deep connection with India's diverse freedom struggle.
- **Value Education:** Teaching courage, honesty, and social service through local examples.
- **Linguistic Heritage:** Promoting reading in the mother tongue.

6. Success Criteria for Students

- The app must allow users to toggle between Kannada and English easily.
- The "Heritage Badge" must be saved in the user's profile.
- The UI must be vibrant and child-friendly.

Project Title :06

Android App Development using GenAI- Kreeda-Ankana (Sports)

1. The Problem Statement:

Many villages have good sports grounds (Volleyball/Cricket), but they are often occupied by the same group all day. There is no system to "Book a Slot" or "Find an Opponent." Small teams have no way to organize a "Friendly Match" with the next village.

2. Detailed Description (The Vision)

Kreeda-Ankana is a "Ground & Match Organizer." It is a digital notice board for the village ground. It allows teams to "Book a Slot" (e.g., 4 PM to 6 PM) and post a "Challenge" to other teams. It turns the village ground into an organized sports hub.

3. App Usage & User Flow

- **Ground Calendar:** See who is playing when.
- **Book Slot:** Reserve a time for your team.
- **Challenge Board:** "Our team is ready for a Volleyball match this Sunday—Who wants to play?"
- **Score Wall:** Post the results of the latest village matches.

4. Technical Implementation & Hints

- **UI:** Use a Grid-based Calendar for time slots.
- **Real-time:** Use Firebase for the challenge board.
- **Database:** Room DB for team profiles and match history.

5. Impact Goals

- **Sports Culture:** Moving sports from casual play to an organized community activity.
- **Youth Harmony:** Using sports to build bridges between different villages/communities.
- **Fitness:** Encouraging more people to participate by making grounds accessible.

6. Success Criteria for Students

- The "Booking" must prevent two teams from picking the same time.
- The "Challenge Board" must allow users to "Reply" to a challenge.
- The UI must be bold and sporty.

Project Title :07

Android App Development using GenAI- Rakta-Vahini (Healthcare)

1. The Problem Statement:

When someone needs blood in a rural hospital, they depend on "Mass Forwards" on WhatsApp. These messages don't reach the "Right Blood Group" in time. There is no "Filter" to find donors by their specific group and "Current Availability."

2. Detailed Description (The Vision)

Rakta-Vahini is a "Filtered Blood Donor Network." It is a privacy-focused emergency directory. Donors register their blood group and "Last Donation Date." When there is a need, the app only shows donors of the "Correct Group" who are "Currently Eligible" (haven't donated in 90 days). It saves time by cutting out the noise.

3. App Usage & User Flow

- **Donor Profile:** Blood Group, Location, and Last Donation date.
- **Emergency Search:** Select group -> See list of eligible donors nearby.
- **I am Eligible:** A toggle for donors to mark themselves as "Ready to Donate."
- **Donation Log:** Keep track of your own donation history.

4. Technical Implementation & Hints

- **Logic:** Calculate eligibility (Current Date - Last Donation Date > 90 days).
- **Location:** Filter results by a 10km/20km radius.
- **Security:** Use Intents to call donors without showing their numbers on the public screen (simulated).

5. Impact Goals

- **Medical Efficiency:** Reducing the time to find blood donors in critical situations.
- **Data Privacy:** Moving from public WhatsApp lists to a managed secure directory.
- **Volunteerism:** Encouraging regular donations through digital tracking

6. Success Criteria for Students

- The "Search" must hide donors who are not currently eligible.
- The app must send a "Thank You" notification after a donation is logged.
- The UI must be professional and life-saving focused.

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Project Title :08

Android App Development using GenAI- Namma-HomeStay (Hospitality)

1. The Problem Statement:

Many houses in Coastal area have extra rooms and serve amazing local food. However, they aren't "Tech-Savvy" enough to register on major hotel booking sites. They miss out on the growing "Eco-Tourism" and "Agri-Tourism" market.

2. Detailed Description (The Vision)

Namma-HomeStay is a "Simplified Host Portal." It is a dedicated app for small rural home-stays. It focuses on the "Local Experience." The host can easily upload photos of their "Today's Menu" (e.g., Akki Rotti, Bamboo shoot curry) and their "Daily Rate." It's designed to be managed by a farmer or a homemaker.

3. App Usage & User Flow

- **My Home Profile:** Photos of the room, toilet, and surrounding farm.
- **Daily Menu:** Update what is special for dinner today.
- **Inquiry Box:** See messages and "Call" buttons from travelers.
- **Local Guide:** List nearby "Secret Spots" (waterfalls, viewpoints).

4. Technical Implementation & Hints

- **UI:** Use CardViews for home-stay listings.
- **Database:** Firebase Firestore for real-time menu/availability updates.
- **Media:** Use ImagePicker for the host to easily update photos.

5. Impact Goals

- **Rural Income:** Providing a third source of income for farming families.
- **Sustainable Tourism:** Promoting low-impact, local-first travel.
- **Digital Literacy: Teaching micro-entrepreneurs how to manage a digital business**

6. Success Criteria for Students

- The "Daily Menu" must be easy to update in 1 minute.
- The "Home Profile" must include a "Verification Checklist" (e.g., Cleanliness).
- The UI must be warm and welcoming.

Project Title : 09

Android App Development using GenAI- Raitha-Varta (Agriculture)

1. The Problem Statement:

Agricultural experts (Krishi Vigyana Kendra) release "Crop Advisories" every week, but these are long PDF files or text messages that farmers find boring. Farmers need "Short, Actionable Tips" specific to their district's soil and weather.

2. Detailed Description (The Vision)

Raitha-Varta is a "Flash-Card Advisor" for farmers. It turns complex research into "Daily 1-Minute Tips." Each card contains one "Action": "Spray this for this pest" or "Best time to weed." It's like "Instagram for Farming," but with scientifically verified information.

3. App Usage & User Flow

- **Daily Tip:** A swipeable card with an image and a 2-sentence instruction.
- **Crop Category:** Filter tips by [Paddy] [Areca nut] [Coconut] [Tomato].
- **Success Story:** A card showing a local farmer who used the tip and got results.
- **Expert Ask:** (Simulated) A button to send a photo of a diseased leaf.

4. Technical Implementation & Hints

- **UI:** Use ViewPager2 for the "Flash-Card" swipe experience.
- **Database:** Room DB to store the daily tips for offline viewing.
- **Media:** Use Glide for high-quality pest/crop images.

5. Impact Goals

- **Precision Farming:** Bringing expert knowledge to the smallest fields.
- **Yield Improvement:** Reducing losses due to pests and incorrect fertilization.
- **Digital Inclusion:** Making scientific data "Digestible" for everyone.

6. Success Criteria for Students

- The app must load the "Daily Tip" instantly on startup.
- The "Flash-Cards" must be available in Kannada.
- The UI must be high-contrast and easy to use in the field.

Project Title : 10

Android App Development using GenAI- Shilpa-Kala (Self-Employment)

1. The Problem Statement:

Wood carvers and "Gombe" makers in places like Channapatna and Kinnala have great skills but struggle with "Product Photography." Their photos on WhatsApp look dull, which makes their high-quality work look "Cheap" to city buyers.

2. Detailed Description (The Vision)

Shilpa-Kala is a "Digital Portfolio Assistant." It is a tool to help artisans "Professionalize" their work. The app includes a "Guided Camera" feature (simulated) that helps them take photos from the right angle. It then adds a "Professional Background" and a "Heritage Label" to the photo, making it ready for a luxury catalog.

3. App Usage & User Flow

- **Product Capture:** A camera interface with an "Outline" to guide the artisan.
- **Brand Overlay:** Automatically adds a "Handmade in Karnataka" logo.
- **Price Tag:** Add name, wood type, and price to the photo.
- **Gallery Share:** Share the "Professional Photo" to WhatsApp or Facebook.

4. Technical Implementation & Hints

- **Camera:** Use CameraX with an "Overlay" (Canvas).
- **Image Processing:** Use a basic Bitmap manipulation to add the logo/text.
- **Storage:** Save the "Branded Photos" in a local folder.

5. Impact Goals

- **Brand India:** Enhancing the "Perceived Value" of Indian handicrafts globally.
- **Digital Literacy:** Teaching artisans the power of "Visual Branding."
- **Economic Empowerment:** Allowing artisans to charge a "Premium Price" for quality-branded work

6. Success Criteria for Students

- The "Overlay" (Logo/Text) must be correctly positioned on the photo.
- The app must allow the artisan to input their name for the "Branding."
- The UI must be clean and focus on the camera.

Project Title : 11

Android App Development using GenAI- Namma-Platform (Infrastructure)

1. The Problem Statement:

In small railway stations, the "Train Announcements" are often inaudible or only in English/Hindi. Rural passengers, especially the elderly, get confused about which platform the "General Coach" will be on.

2. Detailed Description (The Vision)

Namma-Platform is a "Local Station Guide." It is a simplified, Kannada-first railway assistant for small stations. It shows the "Platform Number" and the "Coach Sequence" (Engine -> General -> S1 -> S2). It focuses on the "Next 3 Trains" to avoid clutter and confusion.

3. App Usage & User Flow

- **Live Station:** Select your station.
- **Platform Info:** "Next train: Platform 2."
- **Coach Layout:** A visual strip showing where to stand for the General/Ladies coach.
- **Help Me:** (Simulated) A button to "Speak" the announcement in Kannada.

4. Technical Implementation & Hints

- **UI:** Use a Horizontal ScrollView for the coach layout.
- **Data:** Store the coach positions in a local JSON file.
- **TTS:** Use Text-To-Speech for the Kannada announcements.

5. Impact Goals

- **Citizen Experience:** Making public transport "User-Friendly" for the rural population.
- **Digital Service:** Using tech to solve basic navigation problems at the grassroots level.
- **Inclusion:** Ensuring non-English speakers can use the railways with confidence.

6. Success Criteria for Students

- The "Coach Layout" must be visually clear and match the train name.
- The "Kannada Audio" must be loud and clear.
- The UI must be high-contrast (Blue/Yellow).

Project Title : 12

Android App Development using GenAI- Madhu-Siri (Agriculture)

1. The Problem Statement:

Beekeeping (Apiculture) is a great side-income, but bees are dying due to "Pesticide Spraying" in nearby fields. Beekeepers need to know when their neighbors are spraying, and farmers need to know where the bees are to avoid spraying during the "Active Foraging" hours.

2. Detailed Description (The Vision)

Madhu-Siri is a "Bee-Farmer Harmony" app. It is a communication tool between beekeepers and crop farmers. Beekeepers "Pin" their hive locations. Farmers "Ping" the app when they are about to spray. This allows the beekeeper to "Close the Hives" for 4 hours, saving thousands of bees.

3. App Usage & User Flow

- **Hive Map:** See where the bees are in the village.
- **Spray Alert:** Farmer taps "Spraying Today" -> Beekeepers get a notification**.**
- **Health Check:** A log to track hive health and honey production.
- **Bee-Friendly Tips:** A guide on "Safe Pesticides" that don't harm bees.

4. Technical Implementation & Hints

- **Real-time:** Use Firebase for the "Spray Alerts."
- **Maps:** Use Google Maps API for hive locations.
- **UI:** A simple dashboard for "Bee Health."

5. Impact Goals

- **Biodiversity:** Protecting the pollinators who are vital for food security.
- **Yield Improvement:** Bees increase crop yields by 20-30% through pollination.
- **Sustainable Agriculture:** Promoting "Safe Spraying" practices.

6. Success Criteria for Students

- The "Spray Alert" must reach all "Beekeepers" within a 2km radius.
- Hive locations must be editable by the owner.
- The UI must be clean and use "Nature/Honey" tones.

Project Title : 13

Android App Development using GenAI- Namma-Mela (National Pride)

1. The Problem Statement:

Village Melas (Fairs) have amazing "Drama Troupes" (Company Nataka), but they have no "Booking System." People travel from 50km away only to find the "Front Row" tickets are sold out. They also don't know the "Tonight's Cast" list.

2. Detailed Description (The Vision)

Namma-Mela is a "Drama Company Digital Box-Office." It is a dedicated app for rural theater. It allows the theater manager to post the "Play of the Day" and the "Cast." Fans can "Reserve a Seat" (simulated) and see a "Digital Poster" of the play. It brings professional management to traditional drama.

3. App Usage & User Flow

- **Tonight's Play:** Poster, Name, and Duration of the drama.
- **Cast List:** Photos of the Lead Actor, Comedian, and Singer.
- **Seat Map:** See which "Rows" are still available for booking.
- **Fan Wall:** Leave "Applause" (comments) for the actors after the show.

4. Technical Implementation & Hints

- **UI:** Use a Grid-based Seat Layout (Simulated).
- **Database:** Room DB for persistent seat booking status.
- **Media:** Use Glide for the drama posters.

5. Impact Goals

- **Cultural Economy:** Strengthening the traditional theater industry.
- **Digital Management:** Introducing formal systems to the unorganized art sector.
- **Community Entertainment:** Ensuring rural communities have access to organized arts.

6. Success Criteria for Students

- The "Seat Map" must update when a seat is "Reserved."
- The "Tonight's Play" must be easily updatable by the manager.
- The UI must be "Theatrical" and bold.

Project Title : 14

Android App Development using GenAI- Hasta-Kala Shop (Self-Employment)

1. The Problem Statement:

Artisans making "Small Crafts" (e.g., Banana fiber bags, keychains) struggle with "Inventory." They don't know which color/design is "Best Selling." They keep making everything, leading to "Dead Stock" of items no one wants.

2. Detailed Description (The Vision)

Hasta-Kala Shop is a "Micro-Sales Analytics" tool. It is a very simple "Billing App" for the artisan. Every time they sell an item, they log it. The app then shows a "Pie Chart" (e.g., "60% of your sales are Red Bags"). This data helps the artisan decide "What to make more of."

3. App Usage & User Flow

- **Quick Bill:** Select Item -> Select Color -> Save Sale.
- **Best Seller:** A dashboard showing the "Most Popular" designs.
- **Stock Alert:** "Only 2 Blue Bags left—Time to make more."
- **Income Log:** Total sales for the week/month.

4. Technical Implementation & Hints

- **Visualization:** Use MPAndroidChart for sales analytics (Pie/Bar charts).
- **Database:** Room DB for persistent sales history.
- **UI:** Large icons for different products for fast entry.

5. Impact Goals

- **Business Intelligence:** Giving micro-entrepreneurs the same data power as big retailers.
- **Waste Reduction:** Reducing the production of "Unsold Goods."
- **Growth mindset:** Encouraging artisans to track their "Business Progress."

6. Success Criteria for Students

- The "Best Seller" chart must update instantly after a sale.
- The "Income Log" must allow filtering by "This Week" or "This Month."
- The UI must be intuitive and visual.

Project Title : 15

Android App Development using GenAI - Namma-Railu Buddy (Infrastructure)

1. The Problem Statement:

Local passenger trains are the lifeline for rural-to-city workers. However, platforms are often confusing, and announcements are hard to hear. Passengers often miss their trains or don't know which platform the "General Coach" will be on.

2. Detailed Description (The Vision)

Namma-Railu Buddy is a "Passenger Guide for Local Trains." It is a community-powered platform for small-station passengers. It provides platform numbers, "Coach Position" (Where is the General/Ladies coach?), and crowdsourced delay updates for local passenger trains.

3. App Usage & User Flow

- **Live Station:** Select your station (e.g., Mandya, Birur).
- **Coach Position:** "General Coach is at the front (Engine side)."
- **Platform Ping:** "Train arriving on Platform 2" (User-updated).
- **Alarm:** A "Wake-up Alarm" that triggers when you are 5kms from your destination.

4. Technical Implementation & Hints

- **Location:** Use Geofencing or simple GPS proximity to trigger the destination alarm.
- **Real-time:** Use Firebase for the "Platform Ping" crowdsourcing.
- **UI:** A simple vertical "Coach Layout" visualization.

5. Impact Goals

- **Efficient Transport:** Helping the workforce reach their jobs with less stress.
- **Public Service:** Improving the experience of the common citizen using railways.
- **Inclusion:** Making railway navigation easy for first-time or infrequent travelers.

6. Success Criteria for Students

- The "Destination Alarm" must trigger accurately within a 5km radius.
- The "Platform Ping" must show how many users have confirmed it.
- The UI must be high-contrast and easy to read in a crowded station.

Project Title : 16

Android App Development using GenAI - Grama-Waste Tracker (Environment)

1. The Problem Statement:

Waste collection in villages is irregular. Residents have no way to know when the "Kachara Gaadi" (collection tractor) is coming, leading to trash piles on streets.

2. Detailed Description (The Vision)

Grama-Waste Tracker is a "Smart Collection" app. It provides a live view of the waste collection vehicle. Residents get a notification when the tractor is 2 streets away, and they can report "Blackspots" (illegal dumping) with photos for the Panchayat to see.

3. App Usage & User Flow

- **Live Map:** See the real-time location of the collection tractor.
- **Arrival Alert:** "The tractor will reach your door in 5 minutes."
- **Blackspot Report:** Upload photo + GPS of uncleared garbage.
- **Waste Guide:** Icons showing how to separate Dry and Wet waste.

4. Technical Implementation & Hints

- **Google Maps:** Use Google Maps API for real-time tracking.
- **Real-time DB:** Use Firebase to sync the tractor's coordinates.
- **Notifications:** Implement "Nearby" alerts using background location tracking.

5. Impact Goals

- **Swachh Bharat:** Improving village hygiene through predictable waste collection.
- **Sustainability:** Encouraging waste segregation at the source.
- **Citizen Governance:** Giving people a tool to report and fix local issues.

6. Success Criteria for Students

- The tractor icon must move smoothly on the map in real-time.
- The "Blackspot" reports must show up as pins on a separate Admin map.
- The UI must be clean and available in Kannada.

Project Title : 17

Android App Development using GenAI - Budakattu-Sante (Rural Development)

1. The Problem Statement:

Tribal artisans (Budakattu) produce unique items (wild honey, bamboo crafts) but are exploited by middlemen. They lack direct access to urban buyers and fair trade prices.

2. Detailed Description (The Vision)

Budakattu-Sante is a "Forest-to-Home" marketplace. Designed for tribal cooperatives, it allows a leader to list seasonal forest products. Urban buyers can pre-order, and the app tracks which tribal family supplied what quantity, ensuring direct payment.

3. App Usage & User Flow

- **Forest Catalog:** Browse seasonal items like "Wild Amla" or "Herbal Oils."
- **Pre-Order:** Buyers book items before they are harvested.
- **Supply Log:** Track the source family for every batch of honey/craft.
- **MSP Info:** Display government-approved Minimum Support Prices.

4. Technical Implementation & Hints

- **Offline First:** Use Room DB for inventory logging in zero-network forest areas.
- **Media:** Use high-quality image loading (Glide/Coil) for tribal crafts.
- **Localization:** Use audio-based descriptions for semi-literate tribal users.

5. Impact Goals

- **Tribal Empowerment:** Removing middlemen and increasing forest-dweller income.
- **Vocal for Local:** Bringing unique, organic tribal products to the global market.
- **Economic Inclusion:** Bringing the "Ghats" into the digital economy.

6. Success Criteria for Students

- The app must allow offline data entry that syncs when the network is available.
- The pre-order system must have a "Stock Limit" logic.
- The UI should use earthy tones (Green/Brown) to reflect tribal culture.

Project Title : 18

Android App Development using GenAI - Nimma-Guru (Education)

1. The Problem Statement:

Retired professionals in villages want to "Give Back" by tutoring, but there is no way to connect these "Voluntary Teachers" with students needing after-school help.

2. Detailed Description (The Vision)

Nimma-Guru is a "Community Mentorship" app. It connects retired teachers/engineers (Gurus) with local students. It's a non-commercial directory where mentors list their skills and free hours for weekend classes at community centers.

3. App Usage & User Flow

- **Guru Profile:** List skills (Math, Science, Carpentry) and free hours.
- **Search:** Students find a mentor in their own village/street.
- **Class Calendar:** View upcoming local sessions at the "Samudaya Bhavana."
- **Appreciation:** Students post "Thank You" notes for their Gurus.

4. Technical Implementation & Hints

- **Search:** Implement a "Skill Filter" using ChipGroups.
- **Real-time:** Use Firestore to manage the mentor list.
- **UI:** Clean, educational interface with a "Wall of Fame" for top mentors.

5. Impact Goals

- **Quality Education:** Supplementing school learning with expert community knowledge.
- **Active Aging:** Giving retirees a meaningful way to contribute to nation-building.
- **Knowledge Society:** Creating a culture of "Gyaan-Daan" (Knowledge Sharing).

6. Success Criteria for Students

- The search must be fast and filterable by subject.
- The "Guru Profile" must be easy to edit for elderly users.
- The UI must support both Kannada and English.

Project Title : 19

Android App Development using GenAI - Pashu-Aahar (Agriculture)

1. The Problem Statement:

Small farmers often buy expensive branded cattle feed without knowing how to make "Balanced Feed" at home using local grains, which leads to high costs.

2. Detailed Description (The Vision)

Pashu-Aahar is a "Cattle Nutrition Calculator." The farmer enters the cow breed and current milk yield. The app suggests a "Feed Recipe" using local grains (Maize, Cottonseed) to maximize yield at the lowest cost.

3. App Usage & User Flow

- **Cow Profile:** Enter breed (Jersey, Desi), age, and weight.
- **Recipe Generator:** "Mix 2kg Maize + 1kg Cake" based on yield targets.
- **Cost Tracker:** Compare "Home-made" vs. "Market" feed costs.
- **Veterinary Tips:** Short videos on hygiene and fodder storage.

4. Technical Implementation & Hints

- **Logic:** Implement nutrition math (Protein/Energy ratios) in Kotlin.
- **UI:** Use a Stepper UI for data entry.
- **Charts:** Use MPAndroidChart to show "Cost Savings" over time.

5. Impact Goals

- **Dairy Prosperity:** Increasing the profitability of small milk producers.
- **Self-Reliance:** Reducing dependence on expensive industrial cattle feed.
- **Scientific Farming:** Applying veterinary science to every village cowshed.

6. Success Criteria for Students

- The recipe must change dynamically based on the "Milk Yield Target."
- The app must work offline once the formulas are loaded.
- The UI must use clear icons for different types of fodder.

Project Title : 20

Android App Development using GenAI - Mane-Kelsa (Self-Employment)

1. The Problem Statement:

Domestic workers and gardeners in small towns find work only through "word of mouth." They lose income when their regular employer is away because there is no "Local Job Board."

2. Detailed Description (The Vision)

Mane-Kelsa is a "Hyper-Local Work Directory" (A Digital Naka). Workers list their services and "Daily Availability." Residents can see who is available nearby for a 2-hour or full-day task, reducing unemployment gaps.

3. App Usage & User Flow

- **Worker Profile:** Skill (Cleaning, Gardening), photo, and daily rate.
- **Availability Toggle:** A simple "I am available today" switch.
- **Call Direct:** One-tap calling to the worker.
- **Rating:** Simple "Thumbs Up" from local employers to build trust.

4. Technical Implementation & Hints

- **Real-time:** Use Firebase to sync "Online/Offline" availability status.
- **Intent:** Use ACTION_DIAL for the calling feature.
- **UI:** Large buttons and high-contrast text for semi-literate users.

5. Impact Goals

- **Unorganized Sector Growth:** Digitizing the local labor market for better wages.
- **Economic Security:** Ensuring workers have a steady stream of local opportunities.
- **Urban-Rural Synergy:** Professionalizing services in growing tier-3 towns.

6. Success Criteria for Students

- The "Availability" status must update instantly for all users.
- The worker list must be sorted by "Nearest Street/Area."
- The UI must be localized 100% in Kannada.

Project Title : 21

Android App Development using GenAI - Vidyarthi-Bus (Infrastructure)

1. The Problem Statement:

Students in remote villages depend on specific college buses. If a bus is full or cancelled, they miss exams. They need to know the "Crowd Level" before the bus reaches them.

2. Detailed Description (The Vision)

Vidyarthi-Bus is a "Crowdsourced Bus Alert" app. Students already on the bus report the "Crowd Status" (Empty/Seated/Full). Students waiting at the next stop see this data in real-time to decide if they should wait or find an alternative.

3. App Usage & User Flow

- **Select Route:** Choose your college bus number.
- **Crowd Meter:** See a color-coded status (Green = Seats, Red = Full).
- **Report:** One-tap button: "I'm on the bus, no more seats."
- **Alternative:** View local "Shared Auto" contact numbers.

4. Technical Implementation & Hints

- **Real-time:** Use Firebase Realtime DB for sub-second crowd updates.
- **Logic:** Time-out reports after 15 minutes (to keep data fresh).
- **UI:** Use a Horizontal Progress Bar for the "Crowd Meter."

5. Impact Goals

- **Educational Equity:** Ensuring transport hurdles don't stop rural students from learning.
- **Smart Transit:** Using community data to solve infrastructure gaps.
- **Reliability:** Making rural public transport more predictable

6. Success Criteria for Students

- The "Crowd Meter" must update for all users on the route instantly.
- The app must prevent users who are NOT near the bus route from reporting (Location Check).
- The UI must be lightweight and fast.

Project Title : 22

Android App Development using GenAI - Kumbara-Kala (State Pride)

1. The Problem Statement:

Potters are losing business to plastic/steel. They don't know how to market the "Health Benefits" (like natural cooling or non-toxic cooking) of clay to modern buyers.

2. Detailed Description (The Vision)

Kumbara-Kala is a "Legacy Branding" app. It generates "Digital Story Cards" for clay items (e.g., "This Curd Pot maintains pH balance"). Potters can share these cards on WhatsApp to educate buyers about their village and the science of clay.

3. App Usage & User Flow

- **Catalog:** Photos of pots, lamps, and clay utensils.
- **Story Generator:** Select an item -> Get a "Benefit Card" (Health/Eco tips).
- **Artisan Bio:** "Meet the Maker" – A profile of the potter's heritage.
- **WhatsApp Share:** Send the story card directly to customers.

4. Technical Implementation & Hints

- **Bitmaps:** Use Android's Canvas to overlay text on product images.
- **Share Intent:** Implement WhatsApp sharing for the generated images.
- **UI:** Elegant, artistic interface using traditional textures.

5. Impact Goals

- **Handicraft Revival:** Rebranding traditional skills for the modern health-conscious market.
- **Eco-Friendly Living:** Promoting clay as a plastic alternative.
- **Digital Branding:** Teaching rural artisans the power of "Visual Marketing."

6. Success Criteria for Students

- The "Benefit Card" must be high-quality and readable.
- The app must allow the artisan to add their own name/phone number to the card.
- The UI must feel like a "Gallery."

Project Title : 23

Android App Development using GenAI - Grama-Vaxi (Healthcare)

1. The Problem Statement:

Livestock in villages often die from outbreaks because farmers miss "Vaccination Camp" dates announced only via local loudspeakers.

2. Detailed Description (The Vision)

Grama-Vaxi is a "Livestock Health Alert" app. It tracks the vaccination cycle of every sheep/goat in the village. It acts as a "Digital Health Card" and sends loud notifications 3 days before a government camp reaches the village.

3. App Usage & User Flow

- **Animal Ledger:** Register animals with photo, breed, and age.
- **Vaccine Calendar:** Automatically generates "Next Shot" dates.
- **Camp Alert:** Notification: "Doctor arriving at Temple Square tomorrow."
- **Disease Alert:** Report a sick animal to the local Vet (Simulated).

4. Technical Implementation & Hints

- **WorkManager:** Schedule notifications for months in advance.
- **Room DB:** Store offline animal medical history.
- **UI:** Use large "Animal Icons" for navigation.

5. Impact Goals

- **Livestock Wealth:** Preventing animal loss, which is the "Savings Account" of farmers.
- **Animal Welfare:** Ensuring timely medical care for the rural cattle population.
- **Health Digitization:** Creating a database of village animal health.

6. Success Criteria for Students

- Reminders must trigger even if the app hasn't been opened for days.
- The "Register Animal" form must be simple and visual.
- The UI must support Kannada.

Project Title : 24

Android App Development using GenAI - Sante-Price Index (Micro-Finance)

1. The Problem Statement:

Small vendors in weekly markets (Santes) buy from city Mandis but don't know the "Fair Selling Price." They often sell too cheap (losing profit) or too expensive (losing customers).

2. Detailed Description (The Vision)

Sante-Price Index is a "Vendor's Intelligence" tool. It aggregates city Mandi prices and adds a "Transport + Profit" margin to suggest a "Recommended Retail Price" (RRP) for the village, ensuring the vendor stays competitive and profitable.

3. App Usage & User Flow

- **Price Watch:** See today's Mandi prices for Onion, Tomato, etc.
- **Profit Calc:** Enter your transport cost -> Get suggested selling price.
- **Price Board:** A full-screen "Digital Slate" to show prices to customers.
- **Trends:** See if prices are likely to rise or fall (Simulated).

4. Technical Implementation & Hints

- **Math:** Implement cost-plus pricing algorithms in Kotlin.
- **UI:** "Digital Slate" mode should be high-contrast (Yellow on Black).
- **Firebase:** Fetch (mock) daily prices from a central database.

5. Impact Goals

- **Micro-Entrepreneurship:** Empowering small vendors with data-driven pricing.
- **Consumer Fair-Price:** Ensuring village customers get a fair deal.
- **Financial Literacy:** Teaching vendors how to calculate "Net Profit" vs "Gross Sales."

6. Success Criteria for Students

- The "Profit Calculator" must allow for variable transport/waste costs.
- The "Price Board" must be easy for customers to read from a distance.
- The UI must be rugged and simple.

Project Title : 25

Android App Development using GenAI - Skill-Exchange (Self-Employment)

1. The Problem Statement:

Village technicians (plumbers, electricians) often need each other's help but lack cash to pay. They can "Barter" their skills but don't know who has what skill.

2. Detailed Description (The Vision)

Skill-Exchange is a "Barter Economy" platform for rural technicians. It allows artisans to "Trade Hours." A plumber can fix an electrician's pipes in exchange for motor repair. It builds a self-reliant community of "Skill-Swap" experts.

3. App Usage & User Flow

- **Skill Profile:** "I am an Expert Carpenter."
- **Need Post:** "Need help with leaking roof."
- **Swap Offer:** "I'll do your woodwork in exchange for roof repair."
- **Trust Score:** Based on successful community swaps.

4. Technical Implementation & Hints

- **Firestore:** Use as a "Skill Board" with real-time updates.
- **Logic:** Implement a "Skill Point" system (1 hour = 1 point).
- **UI:** Simple chat-like interface for negotiating swaps.

5. Impact Goals

- **Self-Reliant Communities:** Reducing village dependency on outside technicians.
- **Skill Valorization:** Giving dignity and value to traditional manual skills.
- **Social Capital:** Building trust and cooperation among local workers.

6. Success Criteria for Students

- The "Need Post" must be filterable by "Skill Required."
- The "Trust Score" must increase only after both parties confirm a swap.
- The UI must be community-focused and friendly.

Project Title : 26

Android App Development using GenAI - Namma-Haadi (Infrastructure)

1. The Problem Statement:

Many rural footpaths and shortcuts are not on Google Maps. During monsoons, these paths become flooded, but there is no "Live Status" to warn new travelers or delivery boys.

2. Detailed Description (The Vision)

Namma-Haadi is a "Community Shortcut Guide." It maps the paths "Google forgot." Villagers draw these shortcuts and update their condition daily (Dry/Muddy/Flooded), creating a real-time safety map for the village.

3. App Usage & User Flow

- **Draw Path:** Use GPS to "Trace" a new shortcut on the map.
- **Status Toggle:** Mark a path as "Muddy" or "Flooded" in one tap.
- **Safety Alert:** Warning: "Don't use the forest path after 6 PM."
- **Leaderboard:** Top contributors who keep the map updated.

4. Technical Implementation & Hints

- **Maps:** Use Google Maps SDK Polyline to draw custom paths.
- **Firebase:** Real-time updates for path conditions.
- **Location:** Verify reports by ensuring the user is physically near the path.

5. Impact Goals

- **Last-Mile Connectivity:** Mapping the invisible roads of rural India.
- **Disaster Resilience:** Using community data to navigate monsoon challenges.
- **Digital Mapping:** Empowering villagers to be the "Cartographers" of their own land.

6. Success Criteria for Students

- The "Shortcut" must appear correctly as an overlay on the standard map.
- The "Flooded" status must be visually distinct (e.g., Red/Dashed line).
- The UI must be fast and data-efficient.

Project Title : 27

Android App Development using GenAI-
Namma-Mistri (Self-Employment)

1. The Problem Statement:

Masons (Mistris) are the architects of rural India. However, they lack a way to calculate "Bill of Materials" (Cements, Bricks, Sand) accurately. They often overestimate, leading to wastage, or underestimate, leading to project halts.

2. Detailed Description (The Vision)

Namma-Mistri is a "Construction Assistant" for the local builder. It is a "Site Calculator" in Kannada. The Mistri enters the dimensions of a wall or a room, and the app calculates the number of bricks, bags of cement, and loads of sand required. It also allows the Mistri to maintain a "Daily Wage Log" for their team of laborers.

3. App Usage & User Flow

- **Material Calc:** Enter Length, Width, Height -> Get Material List.
- **Labor Diary:** Record attendance and "Advance Payments" for the team.
- **Site Photos:** A gallery to show "Work Progress" to the house owner.
- **Standard Rates:** A list of local material prices (User-updated).

4. Technical Implementation & Hints

- **Math:** Implement standard civil engineering formulas for volume and area.
- **Database:** Room DB to store multiple "Active Sites" and their logs.
- **UI:** Use a Tabbed Layout (Calculator | Team | Photos).

5. Impact Goals

- **Professionalizing Construction:** Giving rural builders tools similar to large firms.
- **Waste Reduction:** Precision in material estimation saves money and resources.
- **Financial Discipline:** Better tracking of labor wages and advances.

6. Success Criteria for Students

- The "Material Calculator" must allow for different wall thicknesses.
- The "Labor Diary" must calculate the "Balance Due" for each worker automatically.
- The UI must be rugged, with large buttons for use at construction sites.

Project Title : 28

Android App Development using GenAI- Grama-Sanjeevini (Healthcare)

1. The Problem Statement:

In remote villages, the "Local Medical Store" is the only source of advice. If a medicine is out of stock, the villager has to travel 20km to the city. Often, they don't know if the shop in the next village has it.

2. Detailed Description (The Vision)

Grama-Sanjeevini is a "Rural Pharmacy Network." It connects 5-10 village medical stores into a single searchable pool. A villager can search for a medicine, and the app tells them: "Available in [Village B]—3km away." It also allows the pharmacist to set "Expiry Alerts" for their stock to prevent medicine wastage.

3. App Usage & User Flow

- **Medicine Search:** Find availability across nearby shops.
- **Pharmacist Login:** Update stock levels for critical medicines (Insulin, Snake Venom, etc.).
- **Emergency Stock:** A special view for "Life Saving Drugs" availability.
- **Expiry Watch:** Alerts for the pharmacist to sell near-expiry stock at a discount.

4. Technical Implementation & Hints

- **Real-time:** Use Firebase Firestore for a shared inventory view.
- **Location:** Filter shops by a 10km-20km radius.
- **UI:** A very clean, list-based interface.

5. Impact Goals

- **Health Accessibility:** Ensuring critical medicines are found with minimum travel.
- **Supply Chain Efficiency:** Reducing medicine wastage due to expiry.
- **Pharmacist Collaboration:** Turning individual shops into a "Health Network.

6. Success Criteria for Students

- The search must return results from at least 3 different "Mock Shops."
- The "Life Saving" drugs must be highlighted with a red badge.
- The UI must be accessible to people with low digital literacy.

Project Title : 29

Android App Development using GenAI- Ksheera-Sagara (Agriculture)

1. The Problem Statement:

Dairy farmers receive a "Milk Slip" every day showing Fat/SNF content and payment. However, they don't have a way to track their "Monthly Profit" after subtracting the cost of feed, medicines, and labor. Many farmers are actually at a loss without realizing it.

2. Detailed Description (The Vision)

Ksheera-Sagara is a "Dairy Profit/Loss Calculator." It is a dedicated financial tool for the cow-owner. The farmer enters their "Milk Slip" data and their "Expense Data" (Feed bags, Vet visits). The app then shows a clear "Profit per Liter" and suggests how to reduce costs (e.g., switching to home-grown fodder).

3. App Usage & User Flow

- **Income Log:** Enter Liters and Fat% from the daily slip.
- **Expense Log:** Record fodder purchase, electricity, and medicines.
- **Profit Dashboard:** A Green/Red indicator of "Financial Health."
- **Cow-wise Analysis:** (If the farmer has multiple cows) Which cow is most profitable?

4. Technical Implementation & Hints

- **Database:** Room DB to store daily entries for a full year.
- **Analytics:** Use a Pie Chart to show where the most money is being spent (Feed vs Meds).
- **Logic:** Calculate "Net Profit" = (Total Income - Total Expenses).

5. Impact Goals

- **Entrepreneurial Farming:** Moving dairy from a "Survival Activity" to a "Profitable Business."
- **Data-Driven Decisions:** Helping farmers decide when to sell a non-productive cow.
- **Financial Literacy:** Teaching the concept of "Input Cost" vs "Gross Income."

6. Success Criteria for Students

- The app must generate a "Monthly Financial Summary" that can be shared as a PDF/Image.
- The "Expense" entry must have categories like [Fodder] [Medical] [Labor].
- The UI must be clean and focus on "Money" metrics.

Project Title : 30

Android App Development using GenAI- Namma-Skill (Self-Employment)

1. The Problem Statement:

There are many government "Skill Development" centers, but youth in small towns don't know when a "Short-Term Batch" (e.g., 3-month welding or mobile repair) is starting. They miss out on training and remain unemployed.

2. Detailed Description (The Vision)

Namma-Skill is a "Training Opportunity Tracker." It acts as a bridge between Skill Centers and rural youth. The app lists upcoming "Vocational Courses" in the district, the eligibility, and the "Job Guarantee" status. It allows a user to "Apply" with one click, sending their basic profile to the center.

3. App Usage & User Flow

- **Course Finder:** Filter by trade (Electrician, Sewing, Coding).
- **Center Map:** Locate the nearest government skill center.
- **Success Stories: (**Simulated) Photos/Videos of locals who got jobs after the course.
- **Interest Ping:** Tap "I am interested" to get a call back from the trainer.

4. Technical Implementation & Hints

- **UI:** Use ChipGroups for course categories.
- **Database:** Firebase to store a dynamic list of course batches.
- **Notifications:** Alert the user when a new batch starts in their favorite trade.

5. Impact Goals

- **Skill India:** Bridging the gap between training infrastructure and the youth.
- **Unemployment Reduction:** Getting youth into "Job-Ready" courses faster.
- **Equal Opportunity:** Ensuring a village youth has the same info as someone in the city.

6. Success Criteria for Students

- The app must allow filtering by "Duration" (Short term vs Long term).
- The "Apply" button must generate a "Candidate Summary" for the center.
- The UI must be professional and aspirational.

Project Title : 31

Android App Development using GenAI- Karunada-Vanya (National Pride)

1. The Problem Statement:

We have good number of Tigers and Elephants, but people living near forest borders often see wildlife as a "Threat" (Crop raids). There is a need to build "Wildlife Pride" and educate children on the ecological value of these animals.

2. Detailed Description (The Vision)

Karunada-Vanya is a "Wildlife Education & Alert" app. It serves two purposes: "Wonder" and "Safety." It educates children about Karnataka's flora and fauna through interactive cards. For adults, it provides a "Border Alert" system where they can report a "Leopard Sightings" or "Elephant Movements" to alert their neighbors.

3. App Usage & User Flow

- **Wildlife Wiki:** Fact cards about the Black Panther, Hornbill, Sandalwood, etc.
- **Movement Alert:** A "Community Ping" to report wildlife near the village.
- **Co-existence Guide:** "What to do if an elephant enters your field" (Safety tips).
- **Forest Sounds:** (Simulated) Audio clips of different birds and animals.

4. Technical Implementation & Hints

- **UI:** Use MotionLayout for smooth transitions between wildlife cards.
- **Real-time:** Use Firebase for the "Movement Alerts."
- **Media:** Use MediaPlayer for forest sounds.

5. Impact Goals

- **Ecological Balance:** Reducing Human-Wildlife conflict through better information.
- **Nature Conservation:** Inspiring the next generation to protect Karnataka's forests.
- **Sustainable Living:** Promoting the "Co-existence" model of forest-fringe villages

6. Success Criteria for Students

- The "Alert" must have a timestamp and expire after 6 hours (to avoid old data).
- The "Wiki" section must be fully offline-ready.
- The UI must be beautiful, using high-quality wildlife photography.

Project Title : 32

Android App Development using GenAI- Hasta-Shilpa (Self-Employment)

1. The Problem Statement:

Bamboo and Cane artisans in the Western Ghats make incredible furniture and baskets. However, they need "Design Innovation." They keep making the same old patterns, which don't sell well in modern urban markets.

2. Detailed Description (The Vision)

Hasta-Shilpa is a "Design-Bridge for Artisans." It connects traditional skills with modern aesthetics. The app provides a "Design Library" curated for bamboo/cane. It shows "Trending Products" (e.g., Bamboo Laptop Stands, Modern Lamp Shades) and provides "Measurement Guides" so the artisan can replicate the design perfectly.

3. App Usage & User Flow

- **Design Trend:** A feed of modern items made from traditional materials.
- **Blueprint View:** A simple sketch with measurements (e.g., 30cm x 15cm).
- **Material Tracker:** Log how many bamboo poles were used for a batch.
- **Direct Marketplace:** (Simulated) List a finished "Modern Product" for sale.

4. Technical Implementation & Hints

- **UI:** Use CardViews with "Download Blueprint" options (PDF/Image).
- **Storage:** Store high-quality design inspiration photos.
- **Logic:** A "Price Suggester" based on material used + hours worked.

5. Impact Goals

- **Artisanal Modernization:** Keeping traditional crafts relevant in the 21st century.
- **Economic Growth:** Increasing the "Per-Unit Value" of an artisan's work.
- **Sustainable Material:** Promoting Bamboo as a "Green Alternative" to plastic/metal

6. Success Criteria for Students

- The "Blueprint" must be clear and zoomable.
- The "Price Suggester" must allow the artisan to input their local material cost.
- The UI must be clean and focus on "Visual Learning."

Project Title : 33

Android App Development using GenAI- Namma-Raste Health (Infrastructure)

1. The Problem Statement:

Rural roads (PMGSY) are built, but they fail quickly due to "Water Logging" or "Overloaded Trucks." The villagers see the road being destroyed but don't know who the "Contractor" is or how to report a "Minor Crack" before it becomes a major pothole.

2. Detailed Description (The Vision)

Namma-Raste Health is a "Road Maintenance Tracker." Every rural road gets a "Digital Life-Book." The app lists the road name, the contractor's name, and the "Warranty Period." Citizens can report minor damages. It turns the community into "Road Auditors," ensuring that public money spent on roads lasts for years.

3. App Usage & User Flow

- **Road Directory:** Search for your road name.
- **Damage Report:** Take a photo of a crack/clogged drain -> GPS captures location.
- **Contractor Info:** See who built the road and their contact info (Simulated).
- **Success Map:** A list of "Best Maintained Roads" in the Taluka.

4. Technical Implementation & Hints

- **Database:** Room DB to store road details and maintenance logs.
- **Maps:** Show the "Segment" of the road on a map with a Green/Red health line.
- **UI:** Use a Dashboard view for the Taluka road health.

5. Impact Goals

- **Asset Management:** Protecting long-term public investments in infrastructure.
- **Transparency:** Making contractor information public and accessible.
- **Rural Prosperity:** Good roads reduce transport costs for farmers and students.

6. Success Criteria for Students

- The "Damage Report" must be timestamped.
- The "Road Health" must change based on the number of reports in a 1km stretch.
- The UI must be professional and data-driven.

Project Title : 34

Android App Development using GenAI- Grama-Vasathi (Hospitality)

1. The Problem Statement:

Many city dwellers want to experience "Rural Life" (Agri-Tourism), but they are afraid of the lack of "Basic Hygiene" information. Local families have spare rooms but don't know how to "Market" their home-stay or what "Amenities" a city guest expects.

2. Detailed Description (The Vision)

Grama-Vasathi is a "Rural Home-stay Accelerator." It acts as a guide for both the host and the guest. For the villager, it's a "Hospitality School"—giving tips on room setup, hygiene, and food serving. For the guest, it's a "Verified Catalog" of authentic farm-stays where they can live the "Matti-Vasane" (scent of the soil) experience.

3. App Usage & User Flow

- **Host Training:** A checklist of "Guest Readiness" (Clean sheets, Safe water, Western toilet).
- **Farm Experience:** List activities (e.g., "Cow Milking," "Field Plowing," "Local Cooking").
- **Guest Booking:** (Simulated) Simple calendar to reserve a room.
- **Cultural Guide:** "How to greet locals," "What to wear" (for the city guest).

4. Technical Implementation & Hints

- **UI:** Use a Wizard/Stepper for the host training checklist.
- **Database:** Firebase for a searchable directory of farm-stays.
- **Media:** A high-quality gallery for each farm-stay.

5. "Vikshit Bharat" Impact Goals

- **Reverse Migration:** Making villages an attractive economic destination.
- **Cultural Exchange:** Bridging the Urban-Rural divide through shared living.
- **Income Diversification:** Providing farmers with "Non-Agri" income during lean seasons.

6. Success Criteria for Students

- The "Host Readiness Score" must be calculated based on the checklist.
- The search must allow filtering by "Activities" (e.g., Birdwatching).
- The UI must be warm, welcoming, and "Homy."

Project Title : 35

Android App Development using GenAI- Jatre-Namma Pride (State Pride)

1. The Problem Statement:

The village "Jatre" (Fair) is the biggest social event in rural area. However, for a visitor or someone from outside the village, it’s impossible to know the "Events Schedule" (Rathotsava, Wrestling, Drama, Cattle Fair) or the "Safety Guidelines."

2. Detailed Description (The Vision)

Jatre-Namma is a "Digital Guide to the Village Fair." It captures the energy of the Jatre in an app. It provides a real-time schedule of religious and cultural events, a "Lost & Found" digital board, and a map of the temporary stalls (sweet food, toys). It ensures the Jatre remains an organized and safe celebration for everyone.

3. App Usage & User Flow

- **Live Schedule:** "Ratha starts at 4 PM," "Wrestling at 6 PM."
- **Lost & Found:** Post a photo of a lost item or person (Digital help desk).
- **Parking & Safety:** Find the designated parking zone and First-Aid post.
- **Cultural Stories:** The "Legend" behind why the Jatre is celebrated.

4. Technical Implementation & Hints

- **Real-time:** Use Firebase for the "Lost & Found" and "Live Schedule."
- **Maps:** A custom map with temporary "Jatre Markers."
- **UI:** Use a vibrant, festive theme with traditional motifs.

5. Impact Goals

- **Community Resilience:** Organizing traditional mass gatherings using modern tech.
- **Cultural Continuity:** Ensuring young people stay connected to their village festivals.
- **Public Safety:** Reducing chaos and improving emergency response at large fairs

6. Success Criteria for Students

- The "Live Schedule" must highlight the "Current Ongoing Event."
- The "Lost & Found" must allow users to "Mark as Resolved."
- The UI must handle high traffic (simulated) and load quickly.

Project Title : 36

Android App Development using GenAI- Devara-Kaadu (Natural Resources)

1. The Problem Statement:

"Devara Kaadu" (Sacred Groves) are ancient forest patches protected by local deities. They are biodiversity hotspots. However, many are being encroached upon or their ecological history is being forgotten by the youth. They are seen as "Religious Spots" but not as "Biological Treasures."

2. Detailed Description (The Vision)

Devara-Kaadu is a "Sacred Grove Sentinel." It bridges Faith and Science. The app documents the unique trees, rare herbs, and water sources inside these groves. It tells the "Divine Legend" of the grove while providing "Scientific Facts" about the species found there. It turns the community into "Guardians of the Green Temple."

3. App Usage & User Flow

- **Grove Directory:** A map of Sacred Groves in the district.
- **Species Scan:** (Simulated) Identify a rare tree or orchid inside the grove.
- **Myth & Legend:** Read the traditional story of the presiding deity.
- **Conservation Alert:** Report if someone is dumping waste or cutting trees.

4. Technical Implementation & Hints

- **UI:** Use Jetpack Compose for a "Nature-Journal" style aesthetic.
- **Location:** Use GPS to "Unlock" the history when the user is near the grove.
- **Storage:** A rich database of native forest species.

5. Impact Goals

- **Biodiversity Conservation:** Protecting rare endemic species through community faith.
- **Climate Resilience:** Sacred groves act as "Micro-Climate" coolers and water rechargers.
- **Cultural Heritage:** Valuing the "Traditional Ecological Knowledge" of rural Karnataka

6. Success Criteria for Students

- The "Grove Directory" must include at least 5 different types of groves (e.g., Kaavu, Bana).
- The app must clearly distinguish between "Traditional Beliefs" and "Scientific Facts."
- The UI must be serene, green, and respectful.

Project Title : 37

Android App Development using GenAI- Jenu-Gumpu (Agriculture/Self-Employment)

1. The Problem Statement:

Tribal and rural honey hunters often sell raw honey to middlemen at very low prices. They lack the knowledge of "Value Add" (filtering, grading, branding) and don't know the retail market price in big cities.

2. Detailed Description (The Vision)

Jenu-Gumpu is a "Honey Producer’s Collective" app. It empowers the honey hunter to become a brand. The app provides a "Honey Grading Guide" (color and moisture checks), tracks the floral source (e.g., Coffee blossom, Wildflower), and allows the group to log their "Total Collective Stock" to negotiate better prices with big retailers.

3. App Usage & User Flow

- **Harvest Log:** Record the date, location, and quantity of honey collected.
- **Grading Tool:** A visual guide to check honey quality (Simulated moisture test).
- **Price Monitor:** Real-time retail price vs. wholesale price.
- **Batch Tracker:** Assign a "Batch ID" to honey jars for traceability.

4. Technical Implementation & Hints

- **UI:** Use CardViews for different honey grades.
- **Database:** Room DB for storing harvest history.
- **Logic:** A "Profit Calculator" that shows earnings after filtering costs.

5. Impact Goals

- **Tribal Empowerment:** Improving the livelihoods of forest-dwelling communities.
- **Organic Growth:** Promoting "Forest-to-Table" chemical-free products.
- **Sustainable Harvest:** Guidelines on how to harvest honey without killing the bee colony

6. Success Criteria for Students

- The app must allow users to categorize honey by "Floral Source."
- The "Collective Stock" must be viewable as a sum of all individual entries.
- The UI must be available in Kannada and use icons for grading.

Project Title : 38

Android App Development using GenAI- Parisara-Cycle (Infrastructure)

1. The Problem Statement:

In growing towns, students and workers want to use cycles for short commutes, but they are afraid of "Road Safety" and don't know where the "Shortest/Safest" inner-village routes are. They also lack a way to report "Cycle Path Blockages."

2. Detailed Description (The Vision)

Parisara-Cycle is a "Green Commuter Guide." It maps out "Cycle-Friendly" routes (low traffic, shaded roads). It acts as a community-driven map where users can mark "Potholes" or "Dangerous Intersections" specifically for two-wheelers. It turns cycling from a "Necessity" into a "Proud Green Choice."

3. App Usage & User Flow

- **Safe-Route Map:** Navigation that avoids high-speed highways.
- **Pit-Stop Finder:** Locations of cycle repair shops and water points.
- **Eco-Stats:** "You saved 500g of CO_2 today by cycling."
- **Buddy System:** Find other students on the same route to cycle together for safety.

4. Technical Implementation & Hints

- **Maps:** Use Google Maps API with "Bicycling" layer (where available).
- **Logic:** Calculate CO_2 savings based on distance (e.g., 1km = 120g CO_2).
- **Real-time:** Use Firebase for the "Buddy System" location sharing.

5. Impact Goals

- **Net Zero Targets:** Promoting carbon-neutral transport in tier-2 and tier-3 towns.
- **Public Health:** Reducing lifestyle diseases through daily physical activity.
- **Safety:** Using data to make roads safer for the most vulnerable users

6. Success Criteria for Students

- The "Eco-Stats" must persist and show a "Monthly Total."
- The map must allow users to "Pin" a danger zone.
- The UI must be lightweight and battery-efficient.

Project Title : 39

Android App Development using GenAI- Togalu-Gombe (State Pride)

1. The Problem Statement:

Leather Puppetry (Togalu Gombeyaata) is one of the oldest visual arts. However, the stories are usually told in "Old Kannada," and the younger generation finds them hard to follow. The art is losing its "Entertainment Value" because of the language and pace gap.

2. Detailed Description (The Vision)

Togalu-Gombe is a "Digital Shadow-Theater" companion. It acts as a "Subtitle and Story Guide" for live performances. When a fan attends a show, the app provides a "Scene-by-Scene" breakdown in simple Kannada/English. It also includes an "Interactive Puppet Gallery" where users can learn about the characters' symbolism.

3. App Usage & User Flow

- **Live Assist:** Select the Play (e.g., Ramayana) -> See current scene summary.
- **Puppet Scan: (Simulated) Point at a puppet to see its name and powers.**
- **Artist Connect:** Book a puppet-making workshop or buy a miniature puppet.
- **History Feed:** Short videos of how the leather is cured and dyed.

4. Technical Implementation & Hints

- **UI:** Use a "Dark Theme" to mimic the shadow-theater aesthetic.
- **Storage:** A rich database of character descriptions and photos.
- **Media:** Use ExoPlayer for traditional musical background clips.

5. Impact Goals

- **Artistic Innovation:** Using technology to bridge the "Generation Gap" in folk arts.
- **Handicraft Support:** Creating a market for miniature puppets as home decor**.**
- **Cultural Literacy:** Ensuring epic stories remain relevant to the digital generation.

6. Success Criteria for Students

- The "Live Assist" must allow users to manually swipe through scenes.
- The "Puppet Gallery" must include zoomable images to show the intricate "perforations."
- The UI must be available in Kannada.

Project Title : 40

Android App Development using GenAI- Nalla-Nudi (Education)

1. The Problem Statement:

Students from Kannada-medium schools often face a "Challenge" when transitioning to English for higher education. They know the concepts but struggle with "Technical Vocabulary" (e.g., Photosynthesis, Trigonometry). Standard dictionaries are too broad and don't help with specific subjects.

2. Detailed Description (The Vision)

Nalla-Nudi is a "Bridge-Dictionary for Technical Terms." It is designed for the transition phase. A student can search for a scientific or mathematical term in English and get a clear, contextual explanation in simple Kannada, along with a "Pronunciation Guide." It's a "Confidence-Builder" app for rural students.

3. App Usage & User Flow

- **Term Search:** Search "Gravity" -> Get "Gurutvakarshane" + Simple Example.
- **Subject Filters:** View terms specific to [Science] [Math] [Commerce].
- **Voice Guide:** Tap to hear how to pronounce the English word correctly.
- **My List:** Save "Difficult Words" for daily revision.

4. Technical Implementation & Hints

- **Database:** Room DB pre-loaded with a technical glossary.
- **TTS:** Use Android Text-To-Speech for the English pronunciation.
- **UI:** Use a clean, educational interface with a "Word of the Day."

5. Impact Goals

- **Equitable Education:** Removing the language barrier for rural students in STEM.
- **Skill Readiness:** Preparing students for technical interviews and exams.
- **Linguistic Pride:** Using the mother tongue as a "Ladder" to learn global languages.

6. Success Criteria for Students

- The search must be instantaneous (under 200ms).
- The app must work 100% offline.
- The "My List" must allow for "Flashcard" style revision.

Project Title : 41

Android App Development using GenAI- Gandha-Siri (Natural Resources)

1. The Problem Statement:

Sandalwood (Srigandha) is synonymous with our culture. While the laws on growing it have been relaxed, small farmers are afraid of "Theft" and "Illegal Smuggling." They lack a way to track their trees and don't know the "Legal Procedure" for harvesting.

2. Detailed Description (The Vision)

Gandha-Siri is a "Sandalwood Farmer's Guard." It is a digital tree register. A farmer can "Tag" every sandalwood tree on their land with a photo and girth measurement. The app provides a "Security Checklist" (fencing, lighting) and a direct "Legal Guide" on how to inform the forest department for harvesting.

3. App Usage & User Flow

- **Tree Register:** Photo, GPS, and Girth of each tree.
- **Growth Tracker:** Graph showing the tree's health over several years.
- **Security Alert:** A "Panic Button" to inform neighbors of suspicious activity nearby.
- **Maturity Calculator:** (Simulated) Estimated time until the heartwood is ready.

4. Technical Implementation & Hints

- **Location:** Use FusedLocationProviderClient for the "Tree Map."
- **Logic:** Calculate "Heartwood Estimate" based on girth and age.
- **Database:** Room DB for persistent tree logs.

5. Impact Goals

- **Natural Wealth:** Reviving "Sandalwood State" status.
- **Safe Investment:** Encouraging farmers to grow high-value timber without fear.
- **Anti-Smuggling:** Using digital records to prove legal ownership of the timber

6. Success Criteria for Students

- The app must generate a unique "Tree ID" for every entry.
- The "Security Alert" must use a simulated SMS or notification system.
- The UI must use a "Sandal/Wood" color palette.

Project Title :42

Android App Development using GenAI- Raksha-Kavach (Healthcare)

1. The Problem Statement:

Construction and factory workers in small towns often neglect "Safety Gear" (PPE) because they find it cumbersome or think they are "Safe enough." There is no visual way to show them the "Risk Level" of their specific task.

2. Detailed Description (The Vision)

Raksha-Kavach is a "Worker Safety Auditor." It is a visual tool used by a site supervisor or the worker themselves. By selecting their "Task" (e.g., Welding, Height Work), the app shows a "3D Avatar" (simulated) wearing the mandatory gear. It also includes a "Daily Safety Quiz" to build a culture of caution.

3. App Usage & User Flow

- **Task Selector:** Choose the job for the day (e.g., "Digging Trench").
- **Safety Gear Checklist:** Helmet, Gloves, Boots, Goggles – with "Check" buttons.
- **Risk Meter:** Shows the "Likely Injury" if gear is ignored (Visual education).
- **Incident Log:** Report "Near Misses" to prevent future accidents.

4. Technical Implementation & Hints

- **UI:** Use Lottie Animations or high-quality icons for safety gear.
- **Database:** Room DB for "Near Miss" logs.
- **Logic:** A "Safety Score" that increases as the worker completes consecutive days without accidents.

5. Impact Goals

- **Occupational Health:** Reducing workplace fatalities in the unorganized sector.
- **Worker Dignity:** Ensuring the safety of the labor force is prioritized.
- **Industrial Safety:** Building a "Safety-First" mindset in India's growing factories

6. Success Criteria for Students

- The "Safety Checklist" must be specific to the task selected.
- The app must send a "Start-of-Day" safety reminder.
- The UI must be high-visibility (Yellow/Black) for industrial use.

Project Title : 43

Android App Development using GenAI- Kabaddi-Arena(Sports)

1. The Problem Statement:

Kabaddi is our heartbeat, but local "Matti" (mud) tournaments lack "Professional Stats." Players don't know their "Success Rate" in raids or tackles, making it hard for them to get noticed by Pro-Kabaddi scouts.

2. Detailed Description (The Vision)

Kabaddi-Arena is a "Personal Performance Scout." It is an analytics tool for the individual player. During or after a match, a friend or coach logs the player's actions (Empty Raid, Touch Point, Super Tackle). The app then generates a "Player Heatmap" (where they get caught) and a "Point Graph."

3. App Usage & User Flow

- **Match Start:** Add the opponent team and date.
- **Live Logger:** Simple "Raid" and "Tackle" buttons to tap during the game.
- **Performance Card:** "Total Points: 10 | Raid Success: 60%."
- **Video Link:** (Simulated) Attach a video of the "Best Raid" to the stats.

4. Technical Implementation & Hints

- **Logic:** Calculate "Success Percentages" dynamically.
- **UI:** Use a "Court Layout" visual to mark where raids happened.
- **Storage:** Save match-wise history in Room DB.

5. Impact Goals

- **Rural Talent:** Digitizing performance data to bridge the gap between village and pro leagues.
- **Discipline:** Encouraging athletes to use data to improve their weak areas.
- **Cultural Continuity:** Modernizing a traditional sport for the next generation.

6. Success Criteria for Students

- The "Live Logger" must allow undoing an accidental tap.
- The "Performance Card" must be shareable as an image (for social media).
- The UI must be fast and handle the rapid pace of a Kabaddi match.

Project Title : 44

Android App Development using GenAI-
Kutira-Kone (Self-Employment)

1. The Problem Statement:

Home-based tailors and boutique owners in villages have "Excess Fabric" (leftovers) that they throw away. At the same time, others are looking for "Small Pieces" for patchwork or doll making. There is no way to trade these scraps.

2. Detailed Description (The Vision)

Kutira-Kone is a "Zero-Waste Fabric Exchange." It is a hyper-local marketplace for the "Leftover Economy." Tailors can list their fabric scraps (Size, Color, Material). Other artisans can "Buy" or "Swap" them. It promotes a circular economy and reduces textile waste in the village.

3. App Usage & User Flow

- **Scrap Upload:** Photo of the fabric + Approximate size (e.g., 0.5 meters).
- **Browse Map:** Find "Silk Scraps" or "Cotton Bits" in your neighborhood.
- **Direct Swap:** A button to request a "Trade" for another material.
- **Design Ideas:** (Simulated) Projects you can make with small scraps (e.g., Masks, Pouches).

4. Technical Implementation & Hints

- **UI:** Use a Grid Layout for the fabric catalog.
- **Database:** Firebase Firestore for real-time local availability.
- **Search:** Filter by "Material Type" (Silk/Cotton/Wool).

5. Impact Goals

- **Sustainability:** Reducing textile landfill waste through "Upcycling."
- **Cost Reduction:** Allowing small artisans to get materials at a fraction of the price.
- **Community Connection:** Encouraging local entrepreneurs to collaborate.

6. Success Criteria for Students

- The "Upload" must require at least one clear photo.
- The "Search" must have a radius filter (e.g., within 5km).
- The UI must be colorful and visual.

Project Title : 45

Android App Development using GenAI- Sahyadri-Siri (Natural Resources)

1. The Problem Statement:

In the Western Ghats (Sahyadri), there are thousands of "Sacred Springs" and "Local Streams" that provide drinking water. However, there is no "Map of Cleanliness." If a stream is polluted upstream, the villagers downstream don't find out until people get sick.

2. Detailed Description (The Vision)

Sahyadri-Siri is a "Community Water-Quality Map." It relies on the "Senses" of the local community. Users report the "Clarity," "Flow," and "Smell" of their local stream. The app aggregates these "Citizen Observations" to create a "Health Map" of the river system. It's an early warning system for water pollution.

3. App Usage & User Flow

- **Stream Report:** Rate Clarity (1-5), Flow (High/Low), and any visible pollution.
- **Health Map:** A blue-to-brown color-coded map of the district's water bodies.
- **Alert Feed:** "Sudden turbidity seen in [Stream X] – Please check before using."
- **Educational Wiki:** Why these streams are vital for the ecosystem.

4. Technical Implementation & Hints

- **Maps:** Use Google Maps API with custom polyline colors for streams.
- **Logic:** Calculate a "Water Health Score" based on user ratings.
- **Location:** Use GPS to verify the report's origin.

5. Impact Goals

- **Water Security:** Protecting the source of drinking water for millions.
- **Disease Prevention:** Early warning of pollution reduces water-borne illnesses.
- **Ecological Pride:** Connecting people with the life-giving nature of the Ghats

6. Success Criteria for Students

- The "Health Map" must update in real-time as reports come in.
- The app must allow "Anonymous Reporting" for illegal industrial dumping.
- The UI must be serene and blue-themed.

Project Title : 46

Android App Development using GenAI- Namma-Shasane (National Pride)

1. The Problem Statement:

Karnataka has the highest number of "Inscriptions" (Shasanas) in India, found on temple walls or village stones. To most people, they are just "Old Rocks." They are being painted over or destroyed because people don't know they are "Historical Documents."

2. Detailed Description (The Vision)

Namma-Shasane is an "Inscription Decoder." It acts as a digital bridge to the past. Users can "Tag" a stone inscription they find. The app provides a (mocked) "Translation" and explains who the king was and what the "Gift" or "Law" was. It turns "Old Stones" into "Talking History."

3. App Usage & User Flow

- **Find a Shasane:** View a map of documented inscriptions nearby.
- **Photo Tag:** Take a photo of an inscription to "Add to the Map."
- **Story View:** Read the translation: "This stone records a land gift by the Rashtrakutas."
- **Preservation Alert:** Report if a historical stone is being damaged.

4. Technical Implementation & Hints

- **UI:** Use a "Parchment/Stone" aesthetic for the story view.
- **Location:** Use GPS coordinates to create a "Heritage Trail."
- **Database:** Store the translations in a local JSON or Room DB.

5. Impact Goals

- **History Preservation:** Protecting the primary sources of Indian history.
- **Tourism Expansion:** Making even the smallest village a "Historical Destination."
- **Cultural Identity:** Strengthening the link between the people and their ancestors.

6. Success Criteria for Students

- The "Map" must show at least 5 "Found" inscriptions.
- The "Translation" must be available in modern Kannada.
- The UI must be respectful and educational.

Project Title : 47

Android App Development using GenAI- Grameen-Light (Energy)

1. The Problem Statement:

Streetlights in villages often remain "ON" during the day due to manual error, wasting electricity, or remain "OFF" for weeks when a bulb is fused because the Panchayat doesn't know about it.

2. Detailed Description (The Vision)

Grameen-Light is a "Citizen-led Streetlight Audit." It allows villagers to report the status of the "Lamp Posts" in their street. It acts as a digital bridge between the resident and the Panchayat office. It helps in "Zero-Dark" nights and ensures no energy is wasted during the day.

3. App Usage & User Flow

- **Pole Map:** See all poles in the village (Simulated).
- **Quick Report:** Tap a pole -> select status (Working/Fused/Burning in Day).
- **Repair Tracker:** See if the complaint has been "Assigned" or "Fixed."
- **Energy Goal:** A visual showing "Energy Saved this month" through daytime reporting.

4. Technical Implementation & Hints

- **Real-time:** Use Firebase to sync the status across the village.
- **UI:** Use a "Dark/Light" theme switch to represent night/day audits.
- **Database:** Room DB for persistent local records.

5. Impact Goals

- **Smart Villages:** Applying IoT concepts manually through community participation.
- **Energy Efficiency:** Reducing waste at the government level.
- **Public Safety:** Ensuring women and children can walk safely at night.

6. Success Criteria for Students

- The "Pole Status" must be visible as a color-coded dot on the map.
- The app must generate a "Complaint ID" for every report.
- The UI must be extremely simple (one-tap reporting).

Project Title : 48

Android App Development using GenAI - Suraksha-Setu (Infrastructure)

1. The Problem Statement:

In rural areas, women and elderly citizens walking alone at night often feel unsafe. Traditional SOS apps notify the police, but in a village, the "First Responders" are usually neighbors or local youth groups.

2. Detailed Description (The Vision)

Suraksha-Setu is a "Hyper-Local Safety Network." It connects the user to a "Circle of Trust" (neighbors, shopkeepers, and volunteers) within a 500m radius. When triggered, it sets off a loud alarm on the phones of nearby "Verified Volunteers" for immediate physical help.

3. App Usage & User Flow

- **Safe-Circle:** Add 5 trusted neighbors/family members.
- **Panic Trigger:** Shake the phone or double-tap power to alert the circle.
- **Live Audio:** Automatically records and streams 30 seconds of audio.
- **Volunteer Mode:** Local youth register as "Verified Responders."

4. Technical Implementation & Hints

- **FCM:** Use Firebase Cloud Messaging for instant, high-priority alerts.
- **Geofencing:** Use Google Geofencing API to alert only those within a 500m radius.
- **Sensors:** Implement SensorManager to detect emergency "Shake" gestures.

5. Impact Goals

- **Women's Safety:** Creating a secure environment for women in rural workforce.
- **Social Cohesion:** Strengthening the "Neighborhood Watch" culture through tech.
- **Emergency Efficiency:** Reducing response time by involving local citizens.

6. Success Criteria for Students

- The "Shake" gesture must trigger the alert even if the app is in the background.
- The alert must include the user's live GPS coordinates.
- The UI must be high-visibility with a "Large SOS Button."

Project Title : 49

Android App Development using GenAI- Grama-Angana (Infrastructure)

1. The Problem Statement:

Community Halls (Anganas/Samudaya Bhavanas) are often empty. Local youth want to use them for sports or training, but they don't know the "Booking Status" or who the "Key Holder" is. This leads to the building falling into decay.

2. Detailed Description (The Vision)

Grama-Angana is a "Community Space Manager." It turns the village hall into a "Hub of Activity." It provides a public calendar where anyone can see when the hall is free for a wedding, a meeting, or a sports practice. It also allows the community to "Crowdfund" for small repairs (e.g., "Need ₹500 for a new fan").

3. App Usage & User Flow

- **Event Calendar:** View "Booked" vs "Free" slots.
- **Request Booking:** A simple form to ask the Panchayat for a slot.
- **Maintenance Jar:** See a list of "Items needed" (Bulbs, Chairs) and pledge support.
- **Event Board:** See what's happening today (e.g., "Health Camp").

4. Technical Implementation & Hints

- **UI:** Use a CalendarView with status markers.
- **Real-time:** Use Firebase for the booking requests.
- **Database:** Room DB to store the list of maintenance items.

5. Impact Goals

- **Asset Utilization:** Ensuring public buildings are used to their full potential.
- **Social Capital:** Bringing people together for shared community goals.
- **Transparency:** Preventing favoritism in the booking of public spaces

6. Success Criteria for Students

- The "Calendar" must prevent "Double Booking" of the same slot.
- The "Maintenance Jar" must show a progress bar for funds/pledges.
- The UI must be professional and community-focused.

Project Title :50

Android App Development using GenAI- Kalavidara-Balaga (Employment)

1. The Problem Statement:

Local musical troupes (Dollu Kunitha, Pooja Kunitha) only get work during the "Wedding Season" or "Festival Season." For the rest of the year, they work as laborers. They have no way to reach "Event Managers" in cities who are looking for authentic performers.

2. Detailed Description (The Vision)

Kalavidara-Balaga is a "Folk Artist Talent Hub." It is a professional marketplace for traditional troupes. Each troupe creates a "Performance Profile"—complete with photos, a list of instruments they use, and their "Service Area." It makes traditional art a "Sustainable Profession" rather than just a hobby.

3. App Usage & User Flow

- **Troupe Profile:** Group photo, Lead contact, and "Art Form" (e.g., Goravara Kunitha).
- **Portfolio Gallery:** Photos and (simulated) video links of performances.
- **Booking Inquiry:** A direct button for event planners to call the troupe.
- **Equipment List:** "We have 10 drums and our own sound system."

4. Technical Implementation & Hints

- **UI:** Use RecyclerView with StaggeredGridLayout for the gallery.
- **Database:** Firebase Firestore for a searchable directory of artists.
- **Integration:** Use Intents for the "Call" feature.

5. Impact Goals

- **Cultural Preservation:** Making folk arts financially viable for the youth.
- **Creative Economy:** Connecting rural talent with the growing urban event market.
- **Soft Power:** Showcasing the diversity of Karnataka's performance arts

6. Success Criteria for Students

- The "Artist Search" must allow filtering by "District" and "Art Type."
- Troupe profiles must load quickly even on slow networks.
- The UI must be vibrant and use traditional textures.

Project Title :51

Android App Development using GenAI- Namma-Yantra Share (Agriculture/Self-Employment)

1. The Problem Statement:

Small farmers cannot afford to buy "Tractors" or "Power Tillers." Large farmers have these machines, but they sit idle for 20 days a month. There is no way for the small farmer to "Rent" these machines easily without a middleman.

2. Detailed Description (The Vision)

Namma-Yantra Share is an "Uber for Tractors." It is a peer-to-peer machinery rental app. Owners can list their machines and their "Hourly Rate." Small farmers can see what is available nearby and "Request" a machine for 4 hours or a day. It’s an efficiency tool for the whole village.

3. App Usage & User Flow

- **Equipment List:** Browse Tractors, Harvesters, or Sprayers nearby.
- **Rental Request:** Select a date and duration -> Send request to owner.
- **Price Calculator:** "4 hours x ₹500 = Total ₹2000."
- **Machine Health:** (Simulated) Last service date and "Condition" rating.

4. Technical Implementation & Hints

- **Real-time:** Use Firebase to show if a machine is "Currently Available."
- **Location:** Show the distance of the machine from the user's field.
- **UI:** Use a Booking Calendar interface.

5. Impact Goals

- **Mechanization:** Bringing machine power to the smallest landholdings.
- **Resource Efficiency:** Ensuring expensive machinery doesn't sit idle.
- **Farmer Income:** Creating a second source of income for machinery owners.

6. Success Criteria for Students

- The "Price Calculator" must handle different rates (Hourly/Daily).
- The "Request" status must be clearly visible (Pending/Accepted/Declined).
- The UI must be rugged and use "Machinery" icons.

Project Title :52

Android App Development using GenAI - Reshme-Namma Pride (Agriculture)

1. The Problem Statement:

Karnataka is the silk capital of India, but silkworms are extremely sensitive to temperature and humidity. A 2-degree change can kill an entire batch of cocoons. Small-scale farmers don't have expensive sensors; they need a way to log and monitor their "Rearing House" environment manually but smartly.

2. Detailed Description (The Vision)

Reshme-Namma is a "Sericulture Guard." It is a digital supervisor for the silkworm rearing house. The farmer enters the temperature and humidity 3 times a day. The app analyzes these against the "Ideal Growth Curve" for that specific stage of the silkworm (Instar) and gives instant advice on whether to increase ventilation or spray water.

3. App Usage & User Flow

- **Batch Tracker:** Start a new batch (date, breed).
- **Climate Entry:** Enter Temp/Humidity from a simple wall thermometer.
- **Smart Advice:** "Temp is high! Open windows and spread wet gunny bags."
- **Coon Harvest Timer:** Alerts the farmer when it's time to transfer to the spinning trays.

4. Technical Implementation & Hints

- **Logic Engine:** Implement a when or if-else structure for the 5 stages of silkworm growth.
- **Visuals:** A color-coded "Climate Dial" (Green-safe, Orange-caution, Red-danger).
- **Database:** Room DB to store batch-wise harvest data.

5. Impact Goals

- **Economic Stability:** Reducing the risk of "Crop Failure" in sericulture.
- **Global Leadership:** Maintaining Karnataka’s position as a premium silk producer.
- **Agri-Tech for All:** Making complex biological monitoring simple for small farmers

6. Success Criteria for Students

- The app must change its "Advice" based on the age (Instar stage) of the silkworms.
- The "Climate Entry" must be restricted to realistic ranges (e.g., 10°C to 45°C).
- The UI must be clear and focus on "Actionable Alerts."

Project Title : 53

Android App Development using GenAI - Namma-Kelsa (Self-Employment)

1. The Problem Statement:

In growing towns, there is a massive demand for "Daily Wage" skilled labor (Painters, Tilers, Gardeners). However, these workers are often exploited by middlemen or remain unemployed because they are not "Discoverable" digitally.

2. Detailed Description (The Vision)

Namma-Kelsa is a "Dignity-First" labor marketplace. It is a local directory where skilled workers create a "Digital Identity." It doesn't just list phone numbers; it shows "Badges of Skill" and "Verified Photos" of their past work. It allows a local homeowner to find a "Painter in 2kms" and hire them directly.

3. App Usage & User Flow

- **Worker Profile:** Photo, Skill Type, Daily Rate, and Location.
- **Work Gallery:** Upload 3 photos of "Recent Work."
- **Availability Toggle: A simple switch:** "Available for Work Today" (On/Off).
- **Customer Search:** Filter by skill and distance.

4. Technical Implementation & Hints

- **Database:** Firebase Firestore to allow real-time availability updates.
- **Storage:** Store worker profile photos and work samples.
- **UI:** Use ChipGroup for selecting skills (e.g., [Plumbing] [Electrical]).

5. Impact Goals

- **Gig Economy Inclusion:** Bringing the unorganized sector into the digital economy.
- **Poverty Alleviation:** Increasing "Working Days" for laborers by making them easier to find.
- **Skill Recognition:** Giving laborers a professional pride through a digital portfolio

6. Success Criteria for Students

- The "Availability Switch" must update the search results instantly.
- The worker profile must have a "Call" button that works with the phone's dialer.
- The UI must be simple enough for a worker to manage their own profile.

Project Title : 54
Android App Development using GenAI – Pratham-Chikitse (Healthcare)

- **The Problem Statement:**

In emergency situations (Choking, Burn, Snake Bite), the first 5 minutes are the most critical. People often panic or follow wrong "Home Remedies" because they don't have access to medically accurate, step-by-step guidance in their local language.

- **Detailed Description (The Vision)**

Pratham-Chikitse is a "First-Aid Emergency Offline Guide." It is a life-saving manual in your pocket. It uses high-quality illustrations and simple Kannada/English text to guide a user through 20 common emergencies. It works without the internet, ensuring it's ready when the user is in a remote area or during a network failure.

- **App Usage & User Flow**

  - **Emergency Tiles:** Large, clear buttons for (Snake Bite, Heart Attack, Fracture, etc.)
  - **Step-by-Step**: 1, 2, 3... simple instructions with "Do's and Don'ts."
  - **Hospital Finder**: A list of the 3 nearest emergency centers (Simulated/Offline).
  - **Audio Mode:** The app reads out instructions so the user can focus on the patient.

- **Technical Implementation & Hints**

  - **Assets:** Store images/illustrations in the drawable folder for offline access.
  - **TTS:** Use Text-To-Speech API to read out instructions in Kannada.
  - **UI:** Use a ViewPager2 for the step-by-step instructions.

- **Impact Goals**

  - **Public Safety:** Reducing preventable deaths through basic medical knowledge.
  - **Rural Resilience:** Empowering villagers to handle emergencies before the ambulance arrives.
  - **Language Equity:** Making medical life-saving info available in the mother tongue

6. Success criteria for students

- The app must load its main content in under 500ms (Critical for emergencies).
- The "Step-by-Step" instructions must be clear and readable under stress.
- The "Audio Mode" must be toggleable with a single large button.

Project Title :55

Android App Development using GenAI – Sahyadri-Samrakshane (Environment)

- **The Problem Statement:**

The Western Ghats are sensitive zones. Local communities often notice forest fires, illegal logging, or landslides, but they don't have a direct way to report these to the forest department with exact coordinates. By the time news reaches officials, the damage is often done.

- **Detailed Description (The Vision)**

Sahyadri-Samrakshane is a "Forest Sentinel" app. It is a citizen-science tool for environmental protection. It allows a trekker or a local villager to report an "Ecological Alert." The app captures the photo and the high-precision GPS location, allowing forest guards to reach the exact spot quickly.

- **App Usage & User Flow**

  - **Alert Type:** Forest Fire, Landslide, Illegal Tree Cutting, Wildlife Sighting.
  - **Photo & GPS:** Automatic capture of the coordinate when the photo is taken.
  - **Alert Status: "Reported," "Verified,"** or **"Team Dispatched."**
  - **Education:** Tips on how to behave in eco-sensitive zones.

- **Technical Implementation & Hints**

  - **CameraX:** For capturing high-res photos.
  - **Location:** Use FusedLocationProviderClient for the most accurate GPS lock.
  - **Database:** Firebase to sync alerts with a central (mocked) dashboard.

- **Impact Goals**

  - **Environmental Security**: Protecting India’s "Water Tower" (The Western Ghats).
  - **Disaster Management**: Early reporting of landslides can save lives.
  - **Community Policing**: Engaging locals in the protection of their own natural resources.

6. Success Criteria for Students

- The app must work even with low signal (cache the report and send when signal returns).
- The GPS coordinates must be displayed on the screen during the report.
- The UI must use "Nature-Friendly" colors (Greens/Earth tones).

Project Title : 56

Android App Development using GenAI- Namma-Santhe Ledger (Finance)

1. The Problem Statement:

Weekly village markets (Santhe) are the lifeblood of rural retail. However, small vendors (selling vegetables, bangles, snacks) struggle to keep track of "Credit" (Udari) given to regular customers. They often lose money because they forget who owes them what.

2. Detailed Description (The Vision)

Namma-Santhe Ledger is a "Simplified Digital Khata" for small vendors. It replaces the messy pocket diary. It is designed for speed—adding a transaction should take 5 seconds. It helps the smallest businessman understand their "Daily Profit" and "Total Pending Dues" at the end of the market day.

3. App Usage & User Flow

- **Quick Entry:** Select Customer -> Enter Amount -> Add "Udari" (Credit).
- **Payment Log:** Record when a customer pays back.
- **Daily Summary:** "Today you sold for ₹2000; Dues pending ₹400."
- **WhatsApp Reminder:** Send a simple text reminder to a customer about their due.

4. Technical Implementation & Hints

- **Database:** Room DB for persistent storage of transactions.
- **Logic:** Use a ViewModel to calculate the "Net Balance" dynamically.
- **UI:** Large numeric keypad for easy amount entry.

5. Impact Goals

- **Financial Inclusion:** Introducing digital bookkeeping to the unorganized retail sector.
- **Micro-Business Stability:** Reducing bad debts for the smallest entrepreneurs.
- **Digital India:** Moving the "Santhe" economy from paper to pixels.

6. Success Criteria for Students

- The app must show the "Total Outstanding" on the home screen.
- The ledger must be searchable by customer name.
- Adding a new transaction must be a 2-step process (Maximum efficiency)

Project Title :57

Android App Development using GenAI- Karunada-Kote Guide (Karnataka Pride)

- **The Problem Statement:**

Karnataka has magnificent forts (Kotes) like Chitradurga or Bidar, but visitors often walk through them without understanding the engineering or the history. Professional guides are expensive or unavailable, and the signboards are often faded.

- **Detailed Description (The Vision)**

Karunada-Kote Guide is a "Virtual Historian." It is a location-aware tour guide. As the user walks through the fort, the app uses simulated "Beacons" or Location Pings to play the history of that specific gate or temple. It uses storytelling to bring the stones to life.

- **App Usage & User Flow**

  - **Fort Selection:** Select the fort you are visiting.
  - **Guided Map:** A map showing the "Must-See" points.
  - **History Audio:** Tap a point on the map to hear its story in Kannada/English.
  - **Photo Challenge:** "Find the 'Onake Obavva' crevice and take a photo" – engaging young tourists.

- **Technical Implementation & Hints**

  - **Media:** Use MediaPlayer to play high-quality audio narration.
  - **State Management:** Track which points the user has "Unlocked" or visited.
  - **UI:** Use a custom-styled Google Map or a high-quality Image Overlay.

- **Impact Goals**

  - **Heritage Tourism:** Increasing the "Time Spent" by tourists at heritage sites.
  - **Educational Tourism:** Making history fun and interactive for students.
  - **Pride in Engineering:** Highlighting the ancient defense and water systems of the forts.

6. Success Criteria for Students

- The audio must continue playing even if the screen is locked.
- The map must accurately show the user's position relative to the fort's landmarks
- The UI should feel "Historic" (Parchment textures/Royal icons).

Project Title :58
Android App Development using GenAI-

Kutira-Kushala (Self-Employment)

1. The Problem Statement:

Cottage industries (Basket weaving, agarbatti rolling, papad making) are often "Invisible" businesses. They lack a way to show their "Production Capacity" to big vendors or bulk buyers, keeping them stuck in a low-income cycle.

2. Detailed Description (The Vision)

Kutira-Kushala is a "Micro-Factory Showcase." It allows home-based businesses to create a professional profile. They can list "What we make," "Daily Capacity" (e.g., 500 baskets/day), and "Bulk Price." It turns a home kitchen or porch into a "Business Entity" that bulk buyers can trust.

3. App Usage & User Flow

- **Business Profile:** Photo of the family/team, Skill area, and Location.
- **Product Catalog:** List items with photos and wholesale prices.
- **Capacity Meter:** "Ready to take orders for 200 units this week" (Manual toggle).
- **Direct Connect:** A button for bulk buyers to call the entrepreneur.

4. Technical Implementation & Hints

- **Database:** Firebase Firestore for a searchable directory of home businesses.
- **UI:** Use CardViews to showcase products.
- **Storage:** Store images of the products and the manufacturing area.

5. Impact Goals

- **Rural Industrialization:** Strengthening the base of the manufacturing pyramid.
- **Women's Empowerment:** Most cottage industries are women-led; this gives them market power.
- **Self-Reliance (Atmanirbhar):** Encouraging local production of daily-use items

6. Success Criteria for Students

- The "Capacity Meter" must be easily updatable by the user.
- The search must allow filtering by "Product Category" (e.g., Food, Craft).
- The UI must be clean and focus on the products.

Project Title :59
Android App Development using GenAI- Namma-Shaale Inventory (Education)

1. The Problem Statement:

Primary/Secondary schools receive sports kits, lab equipment, and tablets, but there is no easy way to track their "Health." Often, a broken tablet or a lost football is only discovered months later. There is no simple way for a teacher to audit the school's assets.

2. Detailed Description (The Vision)

Namma-Shaale Inventory is a "Digital Asset Auditor." It is a simplified asset management tool. Teachers can "Scan" or "Tag" every item in the school. It tracks the "Condition" (Working, Broken, Needs Repair). It ensures that every government-funded resource is used and maintained properly.

3. App Usage & User Flow

- **Asset Register:** Add item (e.g., "Microscope"), Serial Number, and Photo.
- **Condition Update:** Monthly "Health Check" – Green/Yellow/Red status.
- **Issue Log:** "Football lost during match" – with a date and reason.
- **Repair Request:** A list of items that need the SDMC's attention.

4. Technical Implementation & Hints

- **Database:** Room DB to store the full asset list and history of health checks.
- **UI:** Use a Dashboard view showing: "Total Assets: 50 | Needs Repair: 5."
- **Photos:** Use CameraX to document the condition of high-value items.

5. Impact Goals

- **Resource Optimization:** Ensuring taxpayer money spent on school kits is well-tracked.
- **Educational Quality:** Keeping labs and sports rooms functional for students.
- **Accountability:** Building a culture of "Asset Care" within the public school system.

6. Success Criteria for Students

- The "Monthly Health Check" must allow the teacher to update 10 items in under 2 minutes
- The app must generate a simple "Summary Report" that can be shared.
- The UI must be professional and organized.

Project Title : 60

Android App Development using GenAI- Grama-Urja (Energy)

1. The Problem Statement:

In rural areas, power cuts are frequent but unpredictable. Farmers don't know if the power is "On" to run their irrigation pumps without walking all the way to the field. This wastes time and energy.

2. Detailed Description (The Vision)

Grama-Urja is a "Crowdsourced Power Monitor." It uses the community's input to map power availability. If one farmer sees the power has returned, they hit "Power ON." Everyone in that transformer zone gets an alert. It's a simple, human-powered "Smart Grid."

3. App Usage & User Flow

- **Zone Selection:** Select your village/transformer zone.
- **Status Toggle:** "Power is ON" or "Power is OFF" (User-updated).
- **Last Seen:** "Power was seen 10 mins ago."
- **Pump Timer:** A simple tool to calculate how long to run the pump based on crop type.

4. Technical Implementation & Hints

- **Real-time:** Use Firebase Realtime Database for instant status sync.
- **Alerts:** Use Push Notifications when a status changes from OFF to ON.
- **Logic:** Use a timestamp to show how recent the power update is.

5. Impact Goals

- **Productivity:** Saving farmers hours of walking to check power status.
- **Efficient Irrigation:** Helping manage water better by knowing when power is available.
- **Community Intelligence:** Using shared data to solve an infrastructure pain point.

6. Success Criteria for Students

- The status must update on the home screen for all users within 2 seconds.
- The app must show the "Freshness" of the data (e.g., "Updated 2 mins ago").
- The UI must be high-contrast and easy to read outdoors.

Project Title : 61

Android App Development using GenAI- Arogya-Nidhi (Healthcare)

1. The Problem Statement:

Many government health schemes exist, but rural families don't know which one they are eligible for. They often pay out-of-pocket for surgeries because they didn't know a "Free Treatment Card" could have covered it.

2. Detailed Description (The Vision)

Arogya-Nidhi is a "Health Scheme Eligibility Checker." It is a digital counselor. The user enters their family details (Income, Occupation, BPL status), and the app lists the schemes they can apply for. It provides a "Checklist of Documents" needed for each, ensuring the family is ready before they go to the office.

3. App Usage & User Flow

- **Eligibility Quiz:** Answer 5 questions about your family.
- **Scheme Result:** "You are eligible for [Scheme A] and [Scheme B]."
- **Document Guide:** "Keep your Aadhar and BPL card ready."
- **Hospital List:** Find the nearest "Empaneled Hospital" that accepts these schemes.

4. Technical Implementation & Hints

- **Decision Tree:** Implement a complex if-else or Decision Tree logic in Kotlin.
- **Data Storage:** Store the "Document Checklist" locally in a JSON or Room DB.
- **UI:** Use a Stepper UI for the eligibility quiz.

5. Impact Goals

- **Universal Health Coverage:** Ensuring the poorest benefit from government safety nets.
- **Social Justice:** Removing the "Information Barrier" for the most vulnerable.
- **Financial Protection:** Preventing families from falling into debt due to medical costs.

6. Success Criteria for Students

- The "Result" must change accurately based on the income/BPL status input.
- The "Hospital List" must be searchable by District.
- The UI must be empathetic and simple

Project Title : 62
Android App Development using GenAI- Kashta-Kala (Self-Employment)

1. The Problem Statement:

Carpenters and furniture makers in small towns are highly skilled but struggle to show "Modern Designs" to their customers. They usually rely on old catalogs or the customer's imagination,
which often leads to dissatisfaction with the final product.

2. Detailed Description (The Vision)

Kashta-Kala is a "Digital Design Catalog for Carpenters." It is a sales tool for the local artisan. The app comes pre-loaded with "Modern Furniture Designs" (Simulated). The carpenter can show these to the customer, estimate the wood required, and calculate a "Rough Cost" instantly. It professionalizes the local carpentry business.

3. App Usage & User Flow

- **Design Catalog:** Browse high-quality photos of sofas, beds, cabinets.
- **Material Estimator:** Input dimensions->App calculates "Square Feet" of wood needed.
- **Price Quote:** A simple calculator to add labor and material costs.
- **My Portfolio:** A section for the carpenter to add photos of their own finished work.

4. Technical Implementation & Hints

- **Math Logic:** Implement formulas for area and volume calculations.
- **Images:** Use RecyclerView for the design catalog.
- **Persistence:** Save the "Price Quotes" in Room DB for future reference.

5. Impact Goals

- **Micro-Business Tech**: Giving local artisans tools to compete with large furniture brands.
- **Resource Efficiency**: Accurate estimation reduces wood wastage.
- **Customer Trust**:Professional quotes & design options build better business relationships

6. Success Criteria for Students

- The "Material Estimator" must be accurate and handle different wood types.
- The design gallery must allow "Favoriting" designs.
- The UI must be visual-heavy with clean, large images.

Project Title : 63

Android App Development using GenAI- Namma-Hasiru (Natural Resources)

1. The Problem Statement:

"Seed Balls" are a great way to reforest, but people throw them and forget. They don't know if the seeds "Sprouted" or "Died." There is no "Survival Data" for community plantation drives.

2. Detailed Description (The Vision)

Namma-Hasiru is a "Plantation Tracker." It is a tool for "Green Volunteers." When a user plants a seed or a sapling, they "Geo-Tag" it. They are then reminded to "Check Status" after 3 months. It turns a one-time "Plantation Event" into a "Sustainability Mission" with a record of survival.

3. App Usage & User Flow

- **New Plant:** Take a photo + GPS Tag + Species Name.
- **Check-up Reminder:** App alerts you after 90 days to "Update Status."
- **Survival Score:** "Your village has a 70% tree survival rate."
- **Species Guide:** "Best trees for this soil" (based on success data).

4. Technical Implementation & Hints

- **Location:** Use FusedLocationProviderClient for geo-tagging.
- **Alarms:** WorkManager for the 90-day status reminder.
- **Database:** Room DB for persistent tree history.

5. Impact Goals

- **Ecological Stewardship:** Ensuring that "Tree Planting" leads to actual "Tree Growing."
- **Citizen Data:** Creating a database of what species grow best in which region.
- **Net Zero:** Actively contributing to India's carbon sequestration goals.

6. Success Criteria for Students

- The "Tree Map" must show all saplings planted by the community.
- The "Status Update" must allow for a "Growth Photo."
- The UI must be green, earthy, and inspiring.

Project Title : 64

Android App Development using GenAI- Madhu-Marga (Agriculture)

1. The Problem Statement:

Honey bees are vital for pollination, and beekeeping is a great secondary income for farmers. However, bee colonies are sensitive to "Hive Health" (Mites, Temperature). Beginners often lose their colonies because they don't know how to identify the signs of a failing hive.

2. Detailed Description (The Vision)

Madhu-Marga is a "Digital Beekeeper's Diary." It is an AI-guided assistant (simulated) for hive management. The farmer logs "Observations" (e.g., "Queen seen," "Honey flow," "Activity level"). The app analyzes these and provides tips on when to harvest or when to split the colony.

3. App Usage & User Flow

- **Hive Register:** Tag each hive with an ID and location.
- **Inspection Log:** A checklist (Is the Queen present? Any pests seen?).
- **Harvest Tracker:** Log the quantity of honey collected from each hive.
- **Flora Calendar:** A guide on which flowers are blooming nearby to help bees.

4. Technical Implementation & Hints

- **Logic:** Use a Decision Matrix to suggest hive interventions based on user logs.
- **Visualization:** A Progress Bar for the "Honey Flow" season.
- **Database:** Room DB to store hive-wise performance history.

5. Impact Goals

- **Sweet Revolution:** Increasing high-quality honey production in India.
- **Biodiversity:** Bees help in pollination of all surrounding crops, increasing overall yield.
- **Sustainable Income:** Providing farmers with a low-cost, high-value source of income.

6. Success Criteria for Students

- The app must generate an "Intervention Alert" if the user logs "Low Activity."
- The harvest log must show a year-over-year comparison.
- The UI must be clean and use "Honey/Yellow" tones.

Project Title : 65

Android App Development using GenAI- Channapatna-Namma Pride

(National Pride)

- **The Problem Statement:**

The world-famous Channapatna wooden toys face competition from cheap plastic toys. While people love them, they often buy fakes thinking they are real. Artisans need a way to "Verify" their authentic products and tell the "Story of the Wood" to the buyer.

- **Detailed Description (The Vision)**

Channapatna-Namma is an "Authenticity & Storytelling" app. Each authentic toy comes with a unique ID (simulated). When a buyer enters this ID in the app, it shows the "Artisan's Profile" and a video (mocked) of how that toy was made using natural lac dyes. It turns a "Toy" into a "Piece of Heritage."

- **App Usage & User Flow**

  - **Verify My Toy:** Enter the 6-digit ID to see the artisan's photo.
  - **How it's Made:** An educational section on the "Hale Wood" and "Lac" process.
  - **Meet the Maker:** A directory of Channapatna workshops you can visit.
  - **Toy Catalog:** Browse the latest designs (Rocking horses, puzzles, etc.).

- **Technical Implementation & Hints**

  - **Database:** Use Firebase to store the unique Toy IDs and Artisan links.
  - **UI:** Use CardViews for the artisan profiles.
  - **Localization:** Full Kannada support to represent the local craft pride.

- **Impact Goals**

  - **Protecting GI Tags:** Using tech to safeguard "Geographical Indication" products.
  - **Artisan Recognition:** Giving the anonymous maker a name and a face.
  - **Sustainable Play:** Promoting natural, non-toxic toys over plastic ones.

6. **Success Criteria for Student**

- The "Verification" must return the correct Artisan name for a valid ID.
- The "Meet the Maker" map must show at least 5 workshop pins.
- The UI must be vibrant and "Toy-like" (Playful and colorful).

Project Title : 66

Android App Development using GenAI- Grama-Kalyana Sports (Sports)

1. The Problem Statement:

Village-level sports tournaments (Volleyball, Kabaddi) are huge events, but all the scoring is done on paper. There is no way for someone in the next village to see the "Live Score" or for players to have a digital "Career Record" of their wins.

2. Detailed Description (The Vision)

Grama-Kalyana Sports is a "Digital Scoreboard for Local Tournaments." It turns a village match into a "Professional League" experience. A designated "Scorer" updates the app, and everyone in the community can see the live score on their phones. It builds a digital archive of local sports heroes.

3. App Usage & User Flow

- **Tournament Setup:** Add Teams, Players, and the Schedule.
- **Live Scorer:** A simple interface to update points/wickets/goals.
- **Public View:** A "Live" tab for fans to follow the score remotely.
- **Player Stats:** "Man of the Match" tracking and historical career points.

4. Technical Implementation & Hints

- **Real-time:** Use Firebase Realtime Database for live score sync (Low latency).
- **UI:** A bold, high-contrast scoreboard interface.
- **Sharing:** An "Export Scorecard" button to share the final result on WhatsApp.

5. Impact Goals

- **Youth Engagement:** Professionalizing local sports to keep youth motivated.
- **Talent Pipeline:** Creating a digital record that can be shown to state-level scouts.
- **Community Building:** Using technology to bring the village together around sports.

6. Success Criteria for Student

- The "Live Score" must update on the "Fan View" within 1 second.
- The app must allow for at least 3 types of sports (Kabaddi, Volleyball, Cricket).
- The UI must be readable even in bright outdoor sunlight.

Project Title :67

Android App Development using GenAI- Santhe-Connect (Hospitality)

1. The Problem Statement:

Tourists visiting heritage towns often want to experience the local "Santhe" (weekly market) or eat authentic "Khana-Vali" (local mess) food. However, these small establishments aren't on major travel apps, and their timings or locations are only known to locals.

2. Detailed Description (The Vision)

Santhe-Connect is a "Local Flavor Discovery" app. It acts as a digital bridge for the smallest hospitality providers. It maps out weekly markets, local home-stays, and authentic eateries that don't have a digital presence. It helps the "Slow Traveler" find the soul of Karnataka while providing direct income to local families.

3. App Usage & User Flow

- **Local Eatery Map:** Find the nearest "Jolada Rotti" or "Thatte Idli" point.
- **Santhe Calendar:** A schedule of which village has a market on which day.
- **Specialty Tag:** "Buy authentic honey here" or "Best hand-woven towels."
- **Review Wall:** Tourists leave voice notes or photos of their local experience.

4. Technical Implementation & Hints

- **Maps:** Use Google Maps API with custom category icons (Food/Market/Craft).
- **Database:** Firebase Firestore to allow locals to add their own "Pop-up" stalls.
- **Multimedia:** Use Coil for fast image loading of local food menus.

5. Impact Goals

- **Inclusive Tourism:** Spreading tourist spending beyond 5-star hotels in the local economy.
- **Vocal for Local:** Giving digital visibility to micro-entrepreneurs in food and retail space.
- **Cultural Exchange:** Preserving traditional Karnataka hospitality.

6. Success Criteria for Students

- The map must filter locations based on the current day of the week (for Santhes).
- The "Add Location" feature must capture accurate GPS coordinates.
- The UI must be vibrant and use food-related visual cues

Project Title :68

Android App Development using GenAI-
Grama-Khata (Micro-Finance)

1. The Problem Statement:

Small grocery stores in villages still use "Vahi" (physical books) for credit. When a customer loses track or the book gets damaged, it leads to social friction and financial loss. These vendors need a way to send "Digital Reminders" without complicated accounting software.

2. Detailed Description (The Vision)

Grama-Khata is a "Simplified Digital Ledger" for the village shopkeeper. It focuses on the "Trust Relationship." Instead of complex balance sheets, it shows a simple "Give/Take" (Koduvudu/Tegedukolluvudu) interface. It automates the calculation of total debt and allows for one-tap WhatsApp reminders.

3. App Usage & User Flow

- **Customer Profile:** Name and Photo (to avoid confusion between similar names).
- **Transaction Log:** Simple "+" and "-" buttons for credit and payment.
- **Due Dashboard:** A list of people sorted by the highest amount owed.
- **SMS/WhatsApp Alert:** A pre-filled message: "Namaskara, your due at [Shop Name] is ₹[Amount]."

4. Technical Implementation & Hints

- **Database:** Room DB for offline-first data integrity.
- **Intents:** Use Intent.ACTION_SEND to trigger WhatsApp or SMS.
- **Logic:** Use a ViewModel to calculate the "Net Balance" in real-time.

5. Impact Goals

- **Financial Digitization:** Bringing the "Unorganized Credit" system into the digital era.
- **Micro-Enterprise Health:** Reducing the bankruptcy risk for small retailers.
- **Trust Tech:** Using technology to strengthen social bonds in a community.

6. Success Criteria for Students

- The "Net Due" must update instantly after a transaction.
- The app must generate a "Daily Collection Report" in text format.
- The UI must be usable with one hand (for busy shopkeepers).

Project Title :69

Android App Development using GenAI- Namma-Nala (Infrastructure)

1. The Problem Statement:

Minor irrigation canals (Nalas) are the lifeblood of farms. If a canal is blocked by silt or a breach occurs, the farmer at the "Tail-End" gets no water. Currently, reporting a breach involves traveling to the Taluka office, causing critical delays during the crop season.

2. Detailed Description (The Vision)

Namma-Nala is a "Canal Health Monitor." It empowers farmers to act as "Canal Guards." When a farmer spots a leak, a blockage, or illegal water lifting, they report it via the app. It uses GPS and photos to pinpoint the exact location on the canal map, allowing irrigation engineers to respond instantly.

3. App Usage & User Flow

- **Breach Report:** Take a photo of the leak -> GPS captures the location -> Submit.
- **Water Status:** A crowdsourced feed: "Water reached [Village X] today."
- **Maintenance Tracker:** See if your section of the canal is scheduled for cleaning.
- **Silt Alert:** Log areas where heavy silt is reducing water flow.

4. Technical Implementation & Hints

- **Location:** Use FusedLocationProviderClient for precise canal-bank coordinates.
- **Simulation:** Generate mock "Water Level" data for different sections.
- **UI:** A map overlay showing the path of the primary and secondary canals.

5. Impact Goals

- **Water Efficiency:** Ensuring 100% of released water reaches the designated farms.
- **Agricultural Equity:** Protecting the rights of "Tail-End" farmers.
- **Infrastructure Longevity:** Early detection of leaks prevents major canal collapses.

6. Success Criteria for Students

- The report must display the distance from the nearest "Milestone" or village.
- The "Water Status" feed must show a timestamp of the last update.
- The UI must be optimized for low-bandwidth environments.

Project Title :70

Android App Development using GenAI- Yakshagana-Loka (State Pride)

1. The Problem Statement:

Yakshagana is a vibrant art form, but troupes (Melas) struggle to inform fans about their "Tour Schedule." Fans often miss performances because the information is only on local posters or word-of-mouth.

2. Detailed Description (The Vision)

Yakshagana-Loka is a "Digital Stage" for the art of the coast. It is a dedicated hub for fans and artists. It provides a "Live Calendar" of performances across Karnataka. It also includes an "Artist Encyclopedia" where fans can learn about the different "Veshas" (roles) and the legendary performers behind them.

3. App Usage & User Flow

- **Performance Map:** Pins showing where shows are happening tonight.
- **Artist Directory:** Search for your favorite "Bhagavata" (singer) or actor.
- **Talamaddale Radio:** (Simulated) A list of audio clips of famous dialogues.
- **Mela Schedule:** A 30-day itinerary for different troupes.

4. Technical Implementation & Hints

- **Media:** Use ExoPlayer for high-quality audio clips of Yakshagana songs.
- **UI:** Use a CalendarView integrated with a list of events.
- **Firebase:** Use Firestore to allow Mela managers to update their own schedules.

5. Impact Goals

- **Digital Preservation:** Archiving a traditional art form for the younger generation.
- **Artistic Economy:** Increasing audience turnout for traditional performances.
- **Global Reach:** Making a localized art form accessible to the global Kannada diaspora.

6. Success Criteria for Students

- The "Tonight's Shows" list must update based on the current system date.
- The "Artist Profile" must include a gallery of their different Veshas.
- The UI must use the bold, traditional colors of Yakshagana costumes.

Project Title :71

Android App Development using GenAI- Jan-Aushadhi Finder (Healthcare)

1. The Problem Statement:

Generic medicines at Jan-Aushadhi stores are cheaper than branded ones. However, many people don't know where the nearest store is or if the specific medicine they need is available in a "Generic Version."

2. Detailed Description (The Vision)

Jan-Aushadhi Finder is a "Healthcare Savings Tool." It serves two purposes: finding the nearest generic medicine outlet and acting as a "Brand-to-Generic" translator. Users can enter a branded medicine name and see the equivalent salt/generic name along with the potential price difference.

3. App Usage & User Flow

- **Medicine Search:** Enter branded name -> See Generic equivalent and Price.
- **Store Locator:** Map showing Jan-Aushadhi Kendras within 10kms.
- **Stock Request:** (Simulated) Ask a store if they have a medicine in stock.
- **Reminders:** A tool to track monthly medicine refills.

4. Technical Implementation & Hints

- **Search:** Implement a Filterable RecyclerView for a database of 500+ medicine names.
- **Maps:** Use Google Maps API with "Open Now" status (Simulated).
- **Logic:** Calculate "Total Savings" if the user switches to generics for their prescription.

5. Impact Goals

- **Affordable Healthcare:** Reducing out-of-pocket medical expenses for the poor.
- **Health Literacy:** Educating citizens that "Generic" does not mean "Low Quality."
- **Universal Access:** Ensuring government-subsidized medicines reach the last mile.

6. Success Criteria for Students

- The search must be "Fuzzy" (handle small spelling mistakes in medicine names).
- The price comparison must be visually clear (e.g., Branded: ₹100 vs Generic: ₹20).
- The UI must be clean, clinical, and professional.

Project Title :72

Android App Development using GenAI- EV-Grama Charge (Energy)

1. The Problem Statement:

Electric 2-wheelers are becoming popular in small towns. However, "Range Anxiety" is real because there are no dedicated charging stations. Most owners rely on "Friendly Charging"—asking a local shop or a friend to plug in their vehicle.

2. Detailed Description (The Vision)

EV-Grama Charge is a "Community Charging Network." It allows private shop owners or households to list their 15A power sockets as "Public Charging Points." It's an Airbnb-style model for EV charging. A traveler can see who is willing to provide a 1-hour charge for a small fee, making long-distance EV travel possible in rural areas.

3. App Usage & User Flow

- **Point Map:** See shops/homes offering a "Plug-in" service.
- **Booking:** "Reserve" a 1-hour slot at a local Kirana store.
- **Calculator:** See how many kms you can travel with a 30-minute charge (Simulated).
- **Host Profile:** See the type of socket (5A/15A) and the price per hour.

4. Technical Implementation & Hints

- **Real-time:** Use Firebase to show if a charging point is "Currently Busy."
- **Math Logic:** Implement a simple "Charging Speed" calculator based on battery size.
- **UI:** Use Lottie Animations to show a "Battery Charging" visual.

5. Impact Goals

- **Green Mobility:** Accelerating EV adoption in rural India.
- **Micro-Entrepreneurship:** Allowing small shops to earn extra income from "Power Sales."
- **Sustainable Infrastructure:**Building charging grids without massive govt investment.

6. Success Criteria for Students

- The map must show the distance of the host from the user's current location.
- The "Host" must be able to toggle their availability "ON/OFF."
- The UI must use "Electric Blue/Green" themes.

Project Title : 73

Android App Development using GenAI- Namma-Metro Sahaya (Infrastructure)

- **The Problem Statement:**

For people coming from rural areas to Bengaluru, the Metro is intimidating. They struggle with token machines, identifying the right "Interchange" (like Majestic), and knowing which "Exit Gate" leads to the bus stand or hospital.

- **Detailed Description (The Vision)**

Namma-Metro Sahaya is a "First-Timer's Metro Guide." It focuses on the "Human Side" of transit. Instead of just a map, it provides "Visual Directions." It shows photos of the platforms, explains how to tap the card, and provides "End-to-End" guides (e.g., "How to reach NIMHANS from the station").

- **App Usage & User Flow**
    - **Simple Route:** Enter "Where are you?" and "Where to go?"
    - **Visual Guide:** Step-by-step photos: "Follow the Green Line for Silk Institute."
    - **Exit Finder:** "Which gate for KSRTC Bus Stand?"
    - **Fare & Time:** Simple display of ticket price and travel time.

- **Technical Implementation & Hints**
    - **UI:** Use a Stepper UI for the step-by-step visual guide.
    - **Storage:** Store images of station exits locally for offline access.
    - **Logic:** Implement a "Pathfinding" algorithm for the Metro network (Nodes and Edges).

- **Impact Goals**
    - **Urban Inclusion:** Making city infrastructure accessible to rural citizens.
    - **Efficiency:** Reducing crowding at "Help Desks" through digital self-service.
    - **Digital Confidence:** Empowering citizens to use high-tech public systems without fear.

6. Success Criteria for Students

- The app must clearly highlight "Interchange Stations" (Purple/Green line switch).
- The "Visual Guide" must work even if the user has no internet inside the tunnel.
- The UI must be available in large-font Kannada

Project Title : 74

Android App Development using GenAI- Siri-Dhanya Hub(Agriculture)

1. The Problem Statement:

Karnataka is a leader in Millets (Siri-Dhanya). However, small farmers don't know the current "Market Price" in different cities, and consumers don't know how to cook different types of millets (Navane, Sajje, Baragu).

2. Detailed Description (The Vision)

Siri-Dhanya Hub is a "Millet Value Chain" app. It serves both the farmer and the consumer. For the farmer, it provides "Live Mandi Prices" for millets. For the consumer, it is a "Millet Recipe Book" and health guide. It promotes the consumption of climate-resilient crops.

3. App Usage & User Flow

- **Mandi Watch:** Real-time (simulated) prices of millets in Davangere, Bengaluru, etc.
- **Recipe Lab:** Step-by-step Kannada recipes for millet-based dishes.
- **Health Benefits:** "Why Navane is good for Diabetics" – backed by simple facts.
- **Direct Buy:** (Simulated) Connect with a local Farmer Producer Organization (FPO).

4. Technical Implementation & Hints

- **RecyclerView:** Use a complex list to show prices with "Up/Down" trends.
- **Media:** Use Glide for high-quality food photography.
- **Search:** A search bar for recipes based on "Millet Type."

5. Impact Goals

- **Nutrition Security:** Promoting "Superfoods" to fight malnutrition and lifestyle diseases.
- **Climate Action:** Encouraging crops that require 70% less water than paddy.
- **Farmer Prosperity:** Ensuring farmers get the best price by tracking multiple markets.

6. Success Criteria for Students

- The "Mandi Price" must show a "High/Low" for the last 7 days.
- The "Recipe Lab" must allow users to "Save" their favorite dishes.
- The UI must feel "Earthy" and "Natural."

Project Title : 76

Android App Development using GenAI- Kavya-Kanaja (National Pride)

1. The Problem Statement:

Karnataka has a rich literary history, but the younger generation finds it difficult to access these poems or understand the "Old Kannada" meanings. Literary pride is fading among Gen-Z.

2. Detailed Description (The Vision)

Kavya-Kanaja (Poetry Granary) is a "Literary Revival" app. It makes Kannada poetry "Cool" again. It provides a daily poem, audio recitations by famous voices (simulated), and a "Word Meanings" feature. It's like "Duolingo for Kannada Literature," helping users appreciate the beauty of the language.

3. App Usage & User Flow

- **Poem of the Day:** A beautiful card with a famous verse.
- **Listen & Learn:** Play the audio while reading the text.
- **Bhavartha:** A simple explanation of deep philosophical poems.
- **Poet's Corner:** Biographies and famous works of Jnanpith awardees.

4. Technical Implementation & Hints

- **UI:** Use Jetpack Compose to create elegant, book-like typography.
- **Media:** Use MediaPlayer for audio recitations.
- **Data:** Store a collection of 50+ poems in a local JSON file.

5. Impact Goals

- **Cultural Renaissance:** Reconnecting youth with their linguistic roots.
- **Educational Enrichment:** Helping students with their Kannada literature curriculum.
- **Soft Power:** Preserving and promoting one of the world's oldest living languages.

6. Success Criteria for Students

- The "Poem of the Day" must change automatically based on the date.
- The "Word Meanings" must pop up when a user taps a difficult word.
- The UI must be distraction-free and aesthetically pleasing.

Project Title : 76

Android App Development using GenAI- Anthar-Jala Watch (Natural Resources)

1. The Problem Statement:

Groundwater (Anthar-Jala) levels are falling. Most farmers and residents only realize their borewell is failing when it's too late. There is no "Community Map" of water table health to help people plan their water usage.

2. Detailed Description (The Vision)

Anthar-Jala Watch is a "Community Borewell Monitor." It is a crowdsourced water-table map. Users (farmers/homeowners) log the "Depth of Water" or "Yield" of their borewell. The app aggregates this data to show a "Water Stress Heatmap" for the village. It helps the community decide when to stop digging more borewells and start "Recharge" activities.

3. App Usage & User Flow

- **Borewell Log:** Enter depth, year of digging, and current yield (e.g., 2 inches).
- **Water Map:** A color-coded map (Green/Yellow/Red) of the neighborhood's water health.
- **Recharge Guide:** Simple DIY steps for "Borewell Recharge" structures.
- **Alerts:** "The water table in your area has dropped by 10ft this summer."

4. Technical Implementation & Hints

- **Maps:** Use Heatmap Layer in Google Maps API.
- **Logic:** Calculate "Average Depth" for a specific GPS cluster.
- **UI:** Use a Vertical Scale visual to show water depth levels.

5. Impact Goals

- **Water Sustainability:** Moving from "Individual Extraction" to "Community Management."
- **Disaster Prevention:** Predicting borewell failures before the peak of summer.
- **Resource Awareness:** Making the "Invisible" groundwater visible to the public.

6. Success Criteria for Students

- The map must update its "Heat" intensity based on user inputs.
- The "Recharge Guide" must include simple diagrams/sketches.
- The UI must prioritize "Data Accuracy" and "Privacy" (Hide exact house numbers).

Project Title : 77

Android App Development using GenAI - Raitha-Bharosa Hub (Agriculture)

1. The Problem Statement:

Most of the time farmers often rely on traditional intuition for sowing and fertilization. However, changing weather patterns and soil depletion make this intuition less reliable. Many farmers miss the "Optimal Sowing Window"—a 48-hour period where soil moisture and temperature are perfect for a specific crop. Without a data-driven guide, they face reduced yields or wasted seeds. There is a critical need for a localized, simple-to-use tool that translates complex data into actionable farming steps.

2. Detailed Description (The Vision)

Raitha-Bharosa Hub is a "Smart Sowing Assistant" designed for the modern Indian farmer. The app acts as a bridge between atmospheric data and the farmer's field. It isn't just a weather app; it is a decision-support system. A farmer registers their specific plot, selects their crop (e.g., Sugarcane, Ragi, or Paddy), and the app continuously monitors the "Crop Health Velocity." By comparing real-time (or simulated) weather and soil inputs against established agricultural benchmarks, the app tells the farmer exactly what to do "Today"—whether it is sowing, applying fertilizer, or delaying activity due to predicted heavy rain.

3. App Usage & User Flow

- **Onboarding:** The farmer opens the app (available in Kannada/English) and enters their name and primary crop.
- **The Dashboard:** The main screen displays the current "Sowing Index" (a percentage score of how ideal conditions are right now).
- **Input Center:** Farmers can manually log soil test results (N-P-K levels) or current field moisture observations.
- **Action Plan:** A dedicated tab shows a 7-day "Krishi Calendar." If a storm is predicted in 3 days, the app warns the farmer to finish fertilization 24 hours prior.
- **History:** A local log where farmer can see what they did in previous seasons to compare yields.

4. Technical Implementation & Hints

- **UI Framework:** Use **Jetpack Compose** to create a clean, icon-based interface. Since it's for rural use, use large buttons and clear color indicators (Green for 'Go', Red for 'Wait').

- **Data Persistence:** Use **Room Database** to store the farmer's profile, crop history, and the simulated soil parameters.
- **Simulation Logic:** You must create a "Data Generator" class. For example, write a function that generates a random "Moisture Level" between 10% and 40%. Your app logic should then evaluate: if (moisture > 30%) { display "Soil too wet to sow" }.
- **API Integration:** Use **Retrofit** to fetch weather data from an API like OpenWeatherMap. If credits are limited, mock this data using a local JSON file.

5. Impact Goals

- **Precision Farming:** Reducing seed wastage by 15-20% through better timing.
- **Digital Literacy:** Bringing 4th-year engineering logic to the grassroots level to solve food security.
- **Sustainability:** Promoting targeted fertilizer use to prevent soil degradation.

6. Success Criteria for Students

- The app must load the main dashboard within 2 seconds.
- The "Sowing Index" must change dynamically when simulated data changes.
- The UI must be fully accessible in **Local Language**.
- The code must follow the **MVVM architecture** (Model-View-ViewModel).

Project Title :78
Android App Development using GenAI - Arogya-Sahaya Local (Healthcare)

- **The Problem Statement:**

In most of the rural area, healthcare follow-ups are often missed because elderly patients struggle to manage multiple medications and remember the dates of visiting health camps or ASHA worker rounds. This leads to poor health outcomes and preventable complications. There is a need for a "Digital Health Companion" that simplifies schedules for users who may not be tech-savvy.

- **Detailed Description (The Vision)**

Arogya-Sahaya Local is a localized medication and wellness tracker. The app's primary goal is to provide "Zero-Error" schedule for medicine intake & local health checks. It converts complex doctor prescriptions into simple, time-based reminders. Beyond simple alarms, the app tracks vital indicators like blood pressure/glucose levels, creating a local data log that the user can show to health workers during their next visit.

- **App Usage & User Flow**

    - **Medical Profile:** User enters basic info (age, chronic conditions).
    - **Pill Reminder:** A simple interface to add medicine names, dosage, and times (Morning /Afternoon /Night).
    - **ASHA Connect:** A calendar showing dates of local health camps (Student simulates this data).
    - **Vital Log:** A simple form to enter daily BP or Heart Rate.
    - **Emergency Mode:** A large "SOS" button that triggers a simulated call or message.

- **Technical Implementation & Hints**

    - **Alarms:** Use WorkManager or AlarmManager to ensure reminders trigger if the app is closes
    - **Vitals Tracking:** Use Line Chart (e.g., MPAndroidChart library) to visualize health trends over 7 days.
    - **Database:** Use **Room DB** to store medication history and vital logs locally.

- **Impact Goals**

    - **Health Inclusion:** Ensuring, elderly in remote areas have same adherence to medicine as urban dwellers.
    - **Data-Driven Care:** Empowering ASHA workers with organized patient history.
    - **Preventative Health:** Reducing emergency hospitalizations through consistent monitoring.

6. Success Criteria for Students

- **Notifications must trigger accurately even when the device is in "Doze" mode.**
- The Vital Log must successfully generate a 7-day trend graph.
- The UI must use high-contrast colors and large fonts for elderly accessibility.
- The project must implement the Repository Pattern for data handling

Project Title : 79

Android App Development using GenAI - Grama-Suvidha Portal

1. The Problem Statement:

There is often a transparency gap in rural infrastructure. Citizens in a Panchayat are usually unaware of the progress of local works (e.g., building a community hall, pond rejuvenation). This lack of information leads to a lack of community ownership. A platform is needed to visualize "Digital Progress" for village-level assets.

2. Detailed Description (The Vision)

Grama-Suvidha is a transparency-focused project tracker. It allows every villager to stay informed about their community. The app lists all government-funded projects in the local area, showing their budget, expected completion date, and current status. It creates a "Digital Notice Board" that ensures every citizen knows exactly where development stands.

3. App Usage & User Flow

- **Project List**: View all ongoing works (e.g., "Main Road Repair," "Borewell Installation").
- **Progress View**: Tap a project to see a progress bar and status updates (e.g., "50% Complete").
- **Citizen Feedback**: A simple rating system (1-5 stars) and a "Report Issue" button.
- **Language**: Full UI support in Kannada to ensure every villager can participate.

4. Technical Implementation & Hints

- **Live Data:** Use LiveData or Flow to update the project progress bar dynamically.
- **Data Simulation:** Create a mock list of projects in a local JSON file.
- **Images:** Use Coil or Glide to load mock "Before/After" photos of project sites.

5. Impact Goals

- **Good Governance:** Enhancing transparency and accountability at the grassroots level**.**
- **Citizen Engagement:** Moving from passive residents to active "Progress Pilots."
- **Infrastructure Quality:** Peer-monitoring leads to better construction standards.

6. Success Criteria for Students

- The app must support offline viewing of project lists once data is cached.
- The progress bar must accurately reflect the percentage stored in the database.
- The app must allow users to toggle between Kannada and English.
- Students must document their "Mock API" structure.

Project Title : 80

Android App Development using GenAI - Paryavaran-Kavalu (Environment)

1. The Problem Statement:

Illegal garbage dumping in open spaces, known as "Waste Blackspots," poses a significant health risk. Local cleanup volunteers (Swachh Bharat units) often miss these spots because they don't have a centralized reporting system with precise location data.

2. Detailed Description (The Vision)

Paryavaran-Kavalu is a geo-tagging tool for community cleanliness. It empowers citizens to report environmental hazards instantly. When a user sees a waste blackspot, they "Report" it. The app automatically grabs the location and categorizes the waste. This data is visualized on a "Cleanliness Map" for volunteer groups to prioritize cleanup drives.

3. App Usage & User Flow

- **Quick Report**: User taps "New Report," attaches a photo, and selects waste type.
- **Geo-Tagging**: The app automatically logs the current GPS coordinates.
- **Community Map**: A map view showing red pins (Pending) and green pins (Cleaned).
- **Points System**: Users earn "Eco-Karma" points for every verified report.

4. Technical Implementation & Hints

- **Google Maps API**: Integrate a simple MapView to display location pins.
- **Location Services**: Implement FusedLocationProviderClient for GPS coordinates.
- **Data Simulation**: Simulate the "Volunteer Side" by allowing a mock button to "Mark as Cleaned."

5. Impact Goals

- **Swachh Bharat 2.0**: Leveraging technology to maintain a garbage-free India.
- **Environmental Stewardship**: Encouraging youth to take ownership of their surroundings.
- **Public Health**: Reducing disease vectors by clearing illegal dump sites.

6. Success Criteria for Students

- The app must successfully capture and display the user's current Latitude/Longitude.
- Markers on the map must change color based on the status (Reported/Cleaned).
- The photo upload must be compressed to stay under 500KB.
- The code must implement clean separation between UI and Business logic

Project Title : 81

Android App Development using GenAI - Virasat-Namma Guide (Tourism & Travel)

1. The Problem Statement:

Karnataka has thousands of "Hidden Gems"—historical temples and inscriptions that aren't on major tourist maps. Because these sites lack signage, their history is being forgotten, and tourists miss the chance to explore the state's deep heritage.

2. Detailed Description (The Vision)

Virasat-Namma is a "Smart Heritage Guide." It turns a smartphone into a digital historian. The app uses simple location-based discovery to tell users about historical sites nearby. It focuses on storytelling—providing history, architectural significance, and local legends.

3. App Usage & User Flow

- **Site Discovery:** Shows heritage sites within a specific radius (Simulated data).
- **Information Hub:** Rich descriptions in Kannada and English.
- **QR Scanner:** Scan a mock QR code to unlock a "Hidden Fact."
- **Check-in:** A digital "Travel Passport" for site visits.

4. Technical Implementation & Hints

- **Scanning:** Use **Google ML Kit (Barcode Scanning**) for QR functionality.
- **Multimedia:** Use MediaPlayer to play simple history audio clips.
- **Localization:** Use strings.xml for multi-language support.

5. Impact Goals

- **Cultural Preservation:** Documenting oral histories and minor monuments digitally.
- **Local Economy:** Driving tourism to lesser-known villages.
- **Pride in Heritage:** Educating the next generation about Karnataka's civilizational roots.

6. Success Criteria for Students

- The QR scanner must correctly identify the "Site ID" and open the matching page.
- The Audio guide must play/pause without crashing the app.
- The "Check-in" status must be persisted in Room DB.
- The UI must use a theme inspired by Indian temple architecture.

Project Title :82

Android App Development using GenAI - Namma-Raste Reporter (Infrastructure)

1. The Problem Statement:

Potholes and broken streetlights are major causes of accidents. Currently, reporting these issues involves complex paperwork. A "One-Click" infrastructure reporter can streamline maintenance and improve road safety.

2. Detailed Description (The Vision)

Namma-Raste is a rapid-response infrastructure reporting tool. It captures the photo, location, and severity, then assigns a "Ticket ID." It bridges the gap between the citizen on the road and the local maintenance authorities.

3. App Usage & User Flow

- **Capture Mode:** Open camera -> Take Photo -> Select Type (Pothole/Light).
- **Automatic Submission:** App logs time and GPS location automatically.
- **Status Tracker:** User can enter their Ticket ID to see progress.

4. Technical Implementation & Hints

- **Camera API:** Use **CameraX** to capture images.
- **Security:** Implement simple user login to prevent spam.
- **Storage:** Use Room to store reports locally or Firebase.

5. Impact Goals

- **Safe Mobility:** Reducing road accidents through faster infrastructure repairs.
- **Smart Cities/Villages:** Building a digital feedback loop for public works.
- **Efficiency:** Saving government man-hours by providing exact GPS locations of faults.

6. Success Criteria for Students

- A report must be generated and saved in the local DB in under 3 clicks.
- The app must display a unique, non-repeating Ticket ID for every report.
- The camera preview must not lag or distort on mid-range devices.
- The project must follow standard Kotlin naming conventions.

Project Title :83

Android App Development using GenAI - Mahila-Shakti Unnati (Micro-Finance)

1. The Problem Statement:

Self-Help Groups (SHGs) are vital but rely on physical registers. Tracking weekly savings and interest is error-prone, leading to disputes.

2. Detailed Description (The Vision)

Mahila-Shakti Unnati is a digital ledger for women's SHGs. It acts as a "Digital Accountant," tracking contribution history and calculating group capital and loan eligibility.

3. App Usage & User Flow

- **Member Directory:** Add members with photos and details.
- **Savings Entry:** Mark weekly "Paid" or "Pending."
- **Loan Tracker:** Track repayments and interest.

4. Technical Implementation & Hints

- **Database:** Use **Room DB** with relations (One member to many savings entries).
- **Logic**: Use Kotlin to calculate simple interest or group totals.
- **Export**: Allow sharing a summary via WhatsApp (Intent sharing).

5. Impact Goals

- **Women's Empowerment:** Providing rural women with digital tools to manage their own capital.
- **Financial Literacy:** Teaching the basics of credit scores and interest via the app.
- **Transparency:** Eliminating trust issues within community savings groups.

6. Success Criteria for Students

- Total savings must update instantly when a new entry is added.
- The app must prevent a loan from being recorded if the member has an existing unpaid loan.
- Data must be exportable as a clean text string.
- Database migrations must be handled correctly.

Project Title : 84

Android App Development using GenAI - Yatri-Mitra Shared (Mobility)

1. The Problem Statement:

In smaller towns, commuters have no way of knowing when the next shared auto will arrive, leading to wasted time.

2. Detailed Description (The Vision)

Yatri-Mitra is a communal transit tracker. It demonstrates "ETA Logic" by allowing commuters to see simulated vehicles moving on a route and calculating arrival time based on distance.

3. App Usage & User Flow

- **Route Map:** See a line representing a local route.
- **Live Simulation:** Icons representing autos move along the line.
- **ETA Calculator:** App calculates arrival time at the user's stop.

4. Technical Implementation & Hints

- **State Management:** Use StateFlow to update positions.
- **Math Logic: Formula:** ETA = Distance / Speed.
- **UI:** Use a custom Canvas or View for vehicle markers.

5. Impact Goals

- **Optimized Commute:** Reducing waiting times for the rural labor force and students.
- **Urban-Rural Connectivity:** Making shared transport as predictable as city metros.
- **Fuel Efficiency:** Helping drivers plan trips based on where clusters of passengers are waiting.

6. Success Criteria for Students

- Vehicle icons must move smoothly across the screen without "jumping."
- ETA must count down in real-time.
- The app must handle screen rotations without resetting the simulation.
- Code must be documented for the "Simulation Logic" part.

Project Title : 85

Android App Development using GenAI - Halli-Santhe Digital (Retail)

1. The Problem Statement:

Local artisans have limited reach, selling only at weekly markets. Buyers don't know what is available until they travel there.

2. Detailed Description (The Vision)

Halli-Santhe Digital is a "Hyper-Local Marketplace." It serves as a digital catalog, promoting "Vocal for Local" by making goods discoverable digitally.

3. App Usage & User Flow

- **Artisan Upload:** Add Product Name, Price, and 1 Photo.
- **Buyer Browse:** Grid view of products by category.
- **Check Stock:** Send a mock message to the seller.

4. Technical Implementation & Hints

- **UI:** Use a RecyclerView with a GridLayoutManager.
- **Data Storage:** Use Firebase Firestore or Room.
- **Images:** Implement image compression.

5. Impact Goals

- **Economic Growth:** Connecting village artisans directly to urban consumers.
- **Preserving Crafts:** Saving traditional arts (like Channapatna toys) by increasing sales.
- **Digital Commerce:** Moving local markets from purely physical to "Phygital."

6. Success Criteria for Students

- Clicking a product must open a detailed view with a clear "Call to Action."
- Search functionality must work for product names.
- The app must handle "Empty States" gracefully (if no products are uploaded).
- UI should be colorful and vibrant, reflecting a traditional Indian market.

Project Title : 86

Android App Development using GenAI - Jal-Sanchay Tracker (Natural Resources)

1. The Problem Statement:

Many households have rainwater harvesting but no way to track if it's effective. Without data, conservation feels intangible.

2. Detailed Description (The Vision)

Jal-Sanchay Tracker turns rainwater harvesting into a measurable goal. By entering roof area and rainfall, the app calculates "Water Wealth," encouraging sustainable habits.

3. App Usage & User Flow

- **Setup:** User inputs roof area and tank capacity.
- **Data Entry:** Manual entry of rainfall (mm).
- **Dashboard:** Shows "Liters Saved Today" and "Total Savings."
- **Impact Score:** Converts liters into "Household water days."

4. Technical Implementation & Hints

- **Logic: Formula:** Area x Rainfall x 0.0929 x Runoff Coefficient.
- **Visuals:** Use a Progress Bar that mimics a water tank.
- **Database:** Store historical rainfall in Room DB.

5. Impact Goals

- **Water Security:** Helping India meet its water conservation targets at the household level.
- **Sustainability Education:** Making the concept of "Runoff" and "Catchment" real for citizens.
- **Resource Management:** Data-driven gardening and household usage planning.

6. Success Criteria for Students

- The "Water Tank" visual must fill up relative to the data entered.
- The app must provide a monthly report of total water saved.
- Math calculations must be validated (handling non-numeric inputs gracefully).
- The project must include a "Tips" section for better water harvesting.

Project Title : 87

Android App Development using GenAI - Surya-Shakti Solar Monitor (Energy)

1. The Problem Statement:

Many households in rural and semi-urban area are installing rooftop solar panels. However, users often don't understand how much energy they are generating versus consuming. Without a simple monitoring tool, they can't optimize usage (like running a pump during peak sun) to maximize savings.

2. Detailed Description (The Vision)

Surya-Shakti is a personal energy dashboard for solar-enabled homes. It turns raw electrical data into a "Visual Energy Budget." The app allows a user to input daily generation and battery levels to see their "Green Energy Independence" score. It helps the common citizen transition from being a consumer to a "Prosumer" (Producer + Consumer).

3. App Usage & User Flow

- **Generation Log:** Manually enter or simulate the kWh produced by panels today.
- **Consumption Tracker:** Enter current meter readings to calculate net usage.
- **Peak Suggestion:** A notification saying "High Sun: Ideal time for heavy appliances."
- **Savings Report:** A 30-day view of electricity costs saved.

4. Technical Implementation & Hints

- **Visuals:** Use a circular Progress Bar to show the ratio of solar vs. grid usage.
- **Simulation:** Create a function that varies "Daily Generation" based on a mock weather condition (Cloudy/Sunny).
- **Storage:** Use Room DB for persistent daily energy logs.

5. Impact Goals

- **Renewable Transition:** Accelerating the adoption of clean energy at the household level.
- **Energy Security:** Reducing the load on the central power grid.
- **Economic Savings:** Putting money back into the hands of citizens through lower bills.

6. Success Criteria for Students

- The app must accurately calculate "Net Savings" based on a set per-unit rate.
- The UI must be high-contrast (Yellow/Black) for readability in sunlight.
- The logic must handle "Over-generation" scenarios (Export to grid)

Project Title : 88

Android App Development using GenAI - Kreeda-Prerana Scout (Sports)

1. The Problem Statement:

Talented young athletes in rural schools often go unnoticed because there is no structured way to record their physical milestones (sprint times, jump heights). Coaches and scouts cannot identify "Diamonds in the Rough" without a historical data trail of their performance.

2. Detailed Description (The Vision)

Kreeda-Prerana is a grassroots sports talent tracker. It acts as a "Digital Scout" for physical education teachers. The app allows teachers to create profiles for students and log their performance in standard athletic tests. Over time, it creates a "Talent Curve," making it easy to spot students who have the potential to compete at state or national levels.

3. App Usage & User Flow

- **Athlete Profile:** Name, Age, and primary sport (e.g., Kabaddi, Athletics).
- **Trial Logger:** A stopwatch and distance logger for sprints, long jumps, etc.
- **Milestone Badges:** Automatically awards "District Level Ready" badges based on preset benchmarks.
- **Leaderboard:** A simple internal school ranking to boost healthy competition.

4. Technical Implementation & Hints

- **Timer:** Implement a high-precision Chronometer for timing races.
- **Database:** Room DB for storing athlete performance history.
- **Analytics:** Use simple sorting algorithms to generate the leaderboard.

5. Impact Goals

- **Khelo India Support:** Identifying rural talent early to feed into national academies.
- **Physical Literacy:** Encouraging a culture of fitness and record-keeping in schools.
- **Equal Opportunity:** Ensuring a child in a remote village has a digital record of their talent.

6. Success Criteria for Students

- The timer must be accurate to two decimal places.
- The app must allow "Batch Entry" for an entire class of 30 students.
- The "Talent Curve" graph must be clear and easy to interpret.

Project Title : 89

Android App Development using GenAI - Akshara-Deepa Tutor (Education)

1. The Problem Statement:

After-school learning in rural areas is often unguided. While students have textbooks, they lack a way to track their "Chapter Mastery" or identify which subjects they are consistently weak in. This leads to gaps in learning that only surface during final exams.

2. Detailed Description (The Vision)

Akshara-Deepa is a self-study companion for 10th-grade (SSLC) students. It turns the syllabus into a "Mission Map." The app doesn't just provide content; it tracks "Progress Velocity." Students mark chapters as they finish them and take simple "Self-Check" quizzes. The app then highlights the "Gap Areas" that need more attention.

3. App Usage & User Flow

- **Syllabus Tracker:** A checklist of all chapters in Science, Math, and Social Studies.
- **Quiz Mode:** 5-question daily mocks (Simulated data) for each chapter.
- **Strength Map:** A "Spider Web" chart showing which subjects the student has mastered.
- **Daily Goal:** A reminder to complete at least one topic per day.

4. Technical Implementation & Hints

- **UI:** Use a Tree-View or nested RecyclerView to show the subject-chapter hierarchy.
- **Simulation:** Pre-load a local database with 50-100 mock questions.
- **Logic:** Use a scoring algorithm to update the "Strength Map" after every quiz.

5. Impact Goals

- **Quality Education:** Standardizing the self-study process for rural students.
- **Digital Literacy:** Introducing students to data-driven learning early.
- **Improved Results:** Reducing school dropout rates by catching learning gaps early.

6. Success Criteria for Students

- The progress bar must update instantly across the entire subject category.
- The quiz must have a timer and a "Review Answer" section.
- The app must work 100% offline once the syllabus is loaded.

Project Title : 90

Android App Development using GenAI - Karunada-Kala Source (National Pride)

1. The Problem Statement:

Traditional Karnataka art forms like Yakshagana, Kinnala toys, and Bidriware are struggling to reach modern audiences. Many people want to support these art forms but don't know where the authentic training centers or "Guru-Shishya" workshops are located.

2. Detailed Description (The Vision)

Karunada-Kala is a cultural discovery and preservation platform. It serves as a "Directory of Pride." The app maps local artisans and performance troupes across Karnataka. It allows users to find workshops where they can learn these arts, buy authentic products directly from the maker, and see a calendar of upcoming local performances.

3. App Usage & User Flow

- **Art Form Explorer:** A catalog of Karnataka's unique arts with history and videos.
- **Artisan Map:** Locate authentic makers (e.g., Ilkal weavers) on a map.
- **Workshop Sign-up:** A simple form to register interest in learning an art form.
- **Event Feed:** A list of Yakshagana or Dollu Kunitha performances nearby.

4. Technical Implementation & Hints

- **Maps:** Use Google Maps API to pin artisan locations.
- **Media:** Use ExoPlayer or WebView to embed YouTube clips of performances.
- **Storage:** Use Firebase to store a community-contributed list of events.

5. Impact Goals

- **Cultural Sovereignty:** Preserving the unique identity of Karnataka within a globalized world.
- **Creative Economy:** Providing direct market access to rural artisans.
- **Tourism:** Promoting "Cultural Tourism" where people visit villages to learn arts.

6. Success Criteria for Students

- The map must show different icons for "Workshops" vs "Performances."
- The "Artisan Profile" must have a "Tap to Call" feature.
- The UI must use the yellow and red colors of the Karnataka flag tastefully.

Project Title : 91

Android App Development using GenAI - Matru-Sneh Health (Healthcare)

1. The Problem Statement:

Maternal health tracking in rural areas relies heavily on physical cards that are often lost. Expectant mothers need a simple way to track their nutritional intake, fetal movement (kick counts), and their next check-up date without needing high-speed internet.

2. Detailed Description (The Vision)

Matru-Sneh is a "Pocket Pregnancy Guide" designed for rural health. It focuses on the most critical indicators of maternal well-being. The app provides weekly "Baby Growth" updates in Kannada, a simple kick-counter tool, and a nutrition checklist. It serves as a digital backup to the physical Mother-Child health card.

3. App Usage & User Flow

- **Kick Counter:** A large button to tap every time the baby moves, logging the time.
- **Check-up Countdown:** Days remaining for the next Scan/Vaccination.
- **Nutrition Plate:** A checklist of local foods (Ragi, Greens, Pulses) to eat daily.
- **Health Alert:** A red flag system if the user enters "Danger Signs" (e.g., high swelling).

4. Technical Implementation & Hints

- **Data Visualization:** A simple table showing "Kicks per Hour" over a week.
- **Alarms:** WorkManager for vaccination reminders.
- **Database:** Room DB to ensure data is never lost even if the phone reboots.

5. Impact Goals

- **Reduced MMR:** Lowering Maternal Mortality Rates through better monitoring.
- **Healthy Next-Gen:** Ensuring proper nutrition for the mother leads to a healthier child.
- **Health Digitization:** Moving vital records to a digital-first format.

6. Success Criteria for Students

- The kick-counter must prevent "Double Taps" (Debouncing).
- Danger sign alerts must be clearly visible and suggest immediate action.
- The app must be intuitive enough for a first-time smartphone user.

Project Title : 92

Android App Development using GenAI - Kaushalya-Karnataka (Self Employment)

1. The Problem Statement:

Skilled local laborers (electricians, plumbers, carpenters) in small towns don't have a way to showcase their "Verified Work" or fixed price lists. People usually find them through word-of-mouth, which is inconsistent, leading to unemployment periods for the workers.

2. Detailed Description (The Vision)

Kaushalya-Karnataka is a "Skill Showcase" for local entrepreneurs. Unlike massive job portals, this is a hyper-local "Blue-Collar Portfolio." Every worker can list their "Service Cards"—for example, "Fan Repair: ₹200." It builds a "Trust Economy" by allowing neighbors to give "Thumps Up" ratings to local service providers.

3. App Usage & User Flow

- **Worker Portfolio:** Upload photos of previous work (e.g., a finished cabinet).
- **Service List:** Add services with fixed or "Starting at" prices.
- **Hire Me:** A button for customers to request a call (Simulated notification).
- **Review Wall:** Local residents post short text feedback.

4. Technical Implementation & Hints

- **UI:** Use a CardView based layout for the worker's services.
- **Images:** Use ImagePicker to allow workers to document their work.
- **Database:** Use Firebase Firestore to allow customers to see worker profiles in real-time.

5. Impact Goals

- **Entrepreneurship:** Turning laborers into micro-entrepreneurs.
- **Dignity of Labor:** Professionalizing local services through digital profiles.
- **Local Economy:** Keeping money circulating within the local community.

6. Success Criteria for Students

- The "Service Card" must be editable by the worker at any time.
- The search bar must allow filtering by "Category" (e.g., Electrician).
- The app must implement a simple "Rating" logic (Average stars)

Project Title : 93

Android App Development using GenAI - Hasiru-Usiru Mapper (Environment)

1. The Problem Statement:

Urban "Heat Islands" are growing in cities like Bengaluru and Mysuru. There is a lack of data on where "Green Gaps" exist in a neighborhood. Citizens want to plant trees but don't know where they are needed most or what species are native to their specific soil.

2. Detailed Description (The Vision)

Hasiru-Usiru is a community-driven "Green Auditor." It allows citizens to map existing trees in their street and identify "Empty Pits." The app uses a "Tree Census" approach where users tag a tree, identify its species (simulated), and mark its health. This creates a public map that tells the local municipality exactly where new planting is required.

3. App Usage & User Flow

- **Tree Tagger:** Tap location -> Add Tree Photo -> Identify Species.
- **Empty Pit Marker:** A special tag for spots where a tree is missing.
- **Oxygen Score:** A mock calculation showing the oxygen produced by the trees in that street.
- **Species Guide:** Information on native trees (Neem, Honge, Peepal).

4. Technical Implementation & Hints

- **Maps:** Use Marker Clustering on Google Maps to handle many trees in one view.
- **Logic:** Create a mock formula: Tree Girth x Species Factor = Oxygen Score.
- **Database:** Room DB for offline tagging; Firebase for community map sharing.

5. Impact Goals

- **Climate Resilience:** Actively fighting urban heat through data-driven planting.
- **Biodiversity:** Promoting native species over invasive ones.
- **Citizen Science:** Turning every resident into an environmental data scientist.

6. Success Criteria for Students

- The app must allow the user to tag a location within 5 meters of accuracy.
- The species guide must include photos and descriptions in Kannada.
- The "Oxygen Score" must update dynamically as new trees are added to the list.

Project Title : 94

Android App Development using GenAI - Rakta-Seva Connect (Healthcare)

1. The Problem Statement:

In medical emergencies, finding a specific blood group in a local hospital is often a race against time. While blood banks exist, the real-time availability of "Replacement Donors" (people willing to donate immediately) is not organized at the taluka level.

2. Detailed Description (The Vision)

Rakta-Seva Connect is a "Life-Saving Bridge." It is a dedicated, secure directory for voluntary blood donors. Unlike social media groups where requests get lost, this app is a focused "Emergency Alert" system. When a hospital needs blood, a request is pushed to all nearby registered donors of that blood group. It focuses on the "Golden Hour" of medical treatment.

3. App Usage & User Flow

- **Donor Registry:** Sign up with blood group and last donation date.
- **Emergency Request:** A hospital (mocked) or user posts a requirement.
- **Proximity Alert:** Donors within 10km get a "High Priority" notification.
- **Donor Privacy:** Phone numbers are only visible if the donor "Accepts" the request.

4. Technical Implementation & Hints

- **Notifications:** Use FCM (Firebase Cloud Messaging) for instant alerts.
- **Location Logic:** Filter donor lists based on the distance from the hospital.
- **Privacy:** Implement a simple toggle to mark yourself as "Unavailable" for 3 months after a donation.

5. Impact Goals

- **Health Infrastructure:** Strengthening the emergency response ecosystem.
- **Digital Trust:** Creating a verified and secure platform for sensitive medical needs.
- **Volunteerism:** Encouraging a culture of "Blood Donation as a Duty."

6. Success Criteria for Students

- The "Request" must reach the simulated "Donor" phone in under 5 seconds.
- The app must automatically hide donors who have donated in the last 90 days.
- The UI must be clean, urgent, and focused on speed.

Project Title : 95

Android App Development using GenAI - Shale-Namma Pride (Education)

1. The Problem Statement:

Many government schools have improved significantly in terms of facilities (Smart classes, Labs), but parents in villages are often unaware of these upgrades. This leads to parents choosing expensive private schools over the local government school.

2. Detailed Description (The Vision)

Shale-Namma is a "School Transparency & Pride" portal. It allows the school headmaster or SDMC (School Development and Monitoring Committee) to showcase the "Daily Life" of the school. It includes photos of Mid-day meals, lab experiments, and sports wins. It builds a bridge of trust between the school and the parents, making the school a point of community pride.

3. App Usage & User Flow

- **Daily Meal Update:** A photo and menu of the mid-day meal served today.
- **Facility Tour:** A gallery of the school's labs, library, and toilets.
- **Student Stars:** Celebrating the "Student of the Week" or sports winners.
- **Feedback Box:** A way for parents to send suggestions to the committee.

4. Technical Implementation & Hints

- **Gallery:** Use a ViewPager2 for the school's facility tour.
- **Updates:** Use Firebase Realtime Database so parents get live updates.
- **Language:** Every post should have an auto-toggle for Kannada/English.

5. Impact Goals

- **Education for All:** Strengthening the public school system through community oversight.
- **Social Equity:** Ensuring every child, regardless of income, gets a quality education.
- **Digital Governance:** Using data to improve quality of mid-day meals and school facilities.

6. Success Criteria for Students

- The "Daily Meal" section must allow only one update per day
- The "Feedback" must be stored in a way that it is anonymous (if chosen).
- The UI must feel welcoming and "Child-Friendly."

Project Title :96

Android App Development using GenAI - Grama-Sethu (Public Infrastructure)

- **The Problem Statement:**

Small bridges and culverts in rural Karnataka are critical for monsoon connectivity. If a bridge is submerged or damaged, entire villages are cut off. Currently, there is no way for a traveler or local to know the "Crossability Status" of a rural bridge before they reach it.

- **Detailed Description (The Vision)**

Grama-Sethu is a "Rural Connectivity Monitor." It is a crowdsourced status tracker for minor bridges and roads. Local residents (Grama-Kavalu) update the status: "Open," "Submerged," or "Damaged." This information is vital for emergency vehicles, farmers transporting produce, and students traveling to school during the rainy season.

- **App Usage & User Flow**

  - **Bridge Status Map:** Icons showing the "Live" condition of rural crossings.
  - **Quick Report:** "I am at the bridge and it is submerged" – one-tap update.
  - **Alternative Route:** A simple text-based suggestion if a bridge is closed.
  - **Monsoon Alerts:** Push notifications for heavy rainfall in the bridge catchment area.

- **Technical Implementation & Hints**

  - **Status Logic:** Use a Timestamp to show how "Fresh" the report is (e.g., "Updated 10 mins ago").
  - **Maps:** Use custom markers (Green/Yellow/Red) to indicate status on a map.
  - **Simulation:** Generate mock water-level data to trigger a "Submerged" warning.

- **Impact Goals**

  - **Disaster Resilience:** Preventing people from attempting dangerous crossings during floods.
  - **Supply Chain Continuity:** Ensuring rural produce reaches markets via safe routes.
  - **Smart Infrastructure:** Adding a digital layer to physical bridges for better safety.

6. Success Criteria for Students

- The map markers must update for all users within 3 seconds of a report.
- A "Warning" sound should play if a user opens the app near a "Submerged" bridge.
- The UI must be extremely simple—usable even in heavy rain with one hand.

Project Title : 97

Android App Development using GenAI - Namma-Pustaka (Education)

1. The Problem Statement:

In many village schools, the library is just a cupboard of books with no tracking. Students often don't know what books are available, and teachers struggle to keep track of who has borrowed what. This leads to lost books and a lack of reading culture.

2. Detailed Description (The Vision)

Namma-Pustaka is a "Smart Library Assistant" for rural schools. It turns a simple shelf into a digital catalog. Students can browse book covers on the app, see summaries in Kannada, and "Reserve" a book. For the teacher, it acts as a digital register, sending reminders when a book is overdue.

3. App Usage & User Flow

- **Book Catalog:** Browse books by category (Story, Science, History).
- **QR Borrow:** Scan a QR code on the book to "Issue" it to a student profile.
- **Review Corner:** Students can leave a "Star Rating" and a one-sentence review.
- **Reading Leaderboard:** Tracks which student has read the most pages this month.

4. Technical Implementation & Hints

- **Scanning:** Use Google ML Kit for QR code book tracking.
- **Database:** Room DB to manage the Book list and Transaction history.
- **UI:** Use a RecyclerView with a "Grid" layout to show book covers like a digital shelf.

5. Impact Goals

- **Literacy Promotion:** Encouraging a modern reading habit in the digital age.
- **Resource Management:** Protecting and organizing public school assets.
- **Digital Habits:** Teaching children basic "Check-in/Check-out" digital workflows.

6. Success Criteria for Students

- The app must allow adding a new book via a simple camera-based entry.
- The "Overdue" status must turn text color to Red automatically.
- The library catalog must be searchable by Book Name or Author.

Project Title : 98

Android App Development using GenAI - Gokula-Health (Agriculture/Livestock)

1. The Problem Statement:

Small-scale dairy farmers often lose income because they miss the "Heat Cycle" of their cattle or vaccination dates. Livestock health tracking is mostly oral, leading to missed opportunities for breeding and disease prevention.

2. Detailed Description (The Vision)

Gokula-Health is a digital health card for cattle. It helps the farmer manage the "Life Cycle" of their cows. The app tracks milk yield trends, vaccination schedules, and breeding cycles. It serves as a "Health Passport" that increases the value of the cattle when being sold or insured.

3. App Usage & User Flow

- **Cattle Profile:** Register each cow with a photo and "Ear Tag" ID.
- **Milk Diary:** Daily entry of morning/evening milk yield (liters).
- **Vaccination Alert:** Automated reminders for FMD (Foot and Mouth Disease) & other shots.
- **Yield Graph:** A visual chart showing if a cow's milk production is dropping.

4. Technical Implementation & Hints

- **Visualization:** Use MPAndroidChart to plot milk yield over 30 days.
- **Scheduling:** Use AlarmManager for critical medical reminders.
- **Local Storage:** Room DB for storing long-term animal history.

5. Impact Goals

- **White Revolution 2.0:** Increasing dairy productivity through data-driven management.
- **Farmer Income:** Reducing losses by ensuring 100% vaccination coverage.
- **Rural Digitization:** Bringing precision technology to the animal husbandry sector.

6. Success Criteria for Students

- The app must calculate the "Monthly Average Yield" automatically.
- Reminders must work even without internet connectivity.
- The UI must use icons (Cow, Milk, Syringe) to be intuitive for farmers

Project Title :99

Android App Development using GenAI - Shilpa-Kala Showcase (National Pride)

1. The Problem Statement:

Stone carvers (Shilpis) in places like Shivarapatna create world-class idols, but they have no digital presence. International and national buyers don't know how to reach them, and the art is slowly fading as the younger generation sees no "Modern" way to market it.

2. Detailed Description (The Vision)

Shilpa-Kala Showcase is a "Digital Gallery" for traditional stone and wood carvers. It is a portfolio app where a Shilpi can upload photos of their ongoing and finished works. It connects the "Ancient Art" with the "Modern Buyer" by providing a verified platform for these master craftsmen.

3. App Usage & User Flow

- **Artist Portfolio:** High-quality image gallery of sculptures.
- **Work-in-Progress:** A "Timeline" feature showing how a block of stone became a statue.
- **Order Inquiry:** A simple "Enquire" button that sends a pre-formatted WhatsApp message.
- **Heritage Story:** A section explaining history of specific carving style (e.g., Hoysala style).

4. Technical Implementation & Hints

- **Images:** Use Glide for smooth loading of high-resolution sculpture photos.
- **Interaction:** Implement "Zoom" feature for images to show the intricate carving details.
- **Integration:** Use Intents to launch WhatsApp with pre-filled text.

5. Impact Goals

- **Artisanal Empowerment:** Giving rural masters a global window for their skills.
- **Export Potential:** Facilitating "Make in India" for high-value traditional crafts.
- **Heritage Preservation:** Documenting rare carving techniques through digital media.

6. Success Criteria for Students

- The app must handle at least 20 high-res images without slowing down.
- The "Inquiry" button must include the specific product ID in the message.
- The UI must feel like a premium art gallery (Minimalist and clean).

Project Title :100

Android App Development using GenAI - Jeeva-Bindu (Healthcare)

1. The Problem Statement:

In taluka-level hospitals, finding a blood donor for rare groups like O-negative is a crisis. People post on WhatsApp, but those messages often reach the wrong people or reach too late. There is no "Live Directory" of volunteers who are ready to donate in that specific town.

2. Detailed Description (The Vision)

Jeeva-Bindu is a "Rapid Response" blood donor directory. It is a community-owned safety net. Users register as donors, and when there is a local emergency, "Blood Alert" is sent. It focuses on "Location-First" matching to ensure the donor can reach the hospital within the "Golden Hour."

3. App Usage & User Flow

- **Donor Registry:** Group, Age, and Location (Panchayat/Town).
- **Emergency Post:** "Urgent O+ needed at [Hospital Name]" (Visible to all).
- **I'm Coming:** A button for a donor to signal they are on their way, so others don't duplicate effort.
- **Donor Health:**Simple tracker showing when you're eligible to donate again (90-day rule).

4. Technical Implementation & Hints

- **Notifications:** Use Firebase Cloud Messaging (FCM) for urgent alerts.
- **Filter Logic:** Use a List filter to show only "Available" donors for a specific group.
- **Security:** Use a simple "Verify Phone Number" logic (Simulated).

5. Impact Goals

- **Emergency Resilience:** Building a localized, reliable emergency response system.
- **Community Spirit:** Turning the "Blood Donation" into a localized social duty.
- **Zero Delay:** Reducing mortality rates due to blood unavailability in rural areas.

6. Success Criteria for Students

- An "Emergency Alert" must appear on the simulated donor's phone within 5 seconds.
- The app must correctly calculate the "Eligibility Date" after a donation.
- The UI must use the color Red strictly for emergencies and Green for "Ready to Donate."

Project Title : 101

Android App Development using GenAI - Vidya-Vahini (Transport)

1. The Problem Statement:

Students in rural areas often rely on a single state bus or a school van to reach their college. If the bus is canceled or delayed, they wait for hours at lonely stops, often missing their exams or classes.

2. Detailed Description (The Vision)

Vidya-Vahini is a "Student Commute Buddy." It is a crowdsourced status tracker for student-specific routes. If one student sees the bus passing a certain landmark, they "Ping" the app. All other students on that route get an update. It's "Waze for Rural Students."

3. App Usage & User Flow

- **Route Selection:** Choose your College/School route.
- **Bus Ping:** "Bus just crossed the Bridge" – One tap.
- **ETA Display: "**Bus expected at your stop in 12 mins."
- **Safe-Reach:** A button to notify parents/friends once the student reaches college.

4. Technical Implementation & Hints

- **Real-time Updates:** Use Firebase Realtime Database for the "Ping" logic.
- **Time Logic:** Calculate ETA based on the average time between stops (Simulated).
- **Maps:** A simple horizontal "Route Line" showing the bus's last known position.

5. Impact Goals

- **Educational Access:** Ensuring transport uncertainty doesn't lead to school dropouts.
- **Safety:** Reducing the time students (especially girls) spend waiting at isolated stops.
- **Efficiency:** Helping students manage their study time better by knowing commute delays.

6. Success Criteria for Students

- The "Ping" must update the status for all users on that route instantly.
- The app must allow students to "Report a Breakdown" to alert others to find alternatives.
- The UI must be lightweight and work on low-end smartphones.

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